Statistical Regularities in Cocktail Recipes:

An Exploratory Analysis of CocktailDB Adapted from the Framework of Universal Culinary Scaling Laws

Research-style exploratory data analysis report

Dataset: CocktailDB (ID, name, category, ingredients, nutrients, ratings, votes, preparation time, cuisine)

Methodological reference: Bagler, Tewari, Yadav, Singh, Bansal, Dargar, Goel & Sinha, *Universal statistical laws governing culinary design* (arXiv:2604.28021v1).

This report does not reproduce the paper's global-cuisine corpus. It tests which of the paper's statistical questions can be asked, and which patterns actually appear, in a cocktail-specific dataset.

16,895 raw rows · 14,916 unique names · 16,872 recipes in the frequency corpus · 14,466 normalized ingredients

How this study was conducted

This front section documents the computational workflow before any scientific results. It answers three practical questions: how the pipeline was built, what exploratory and preprocessing operations were applied to CocktailDB, and which libraries implemented each step. Later sections report findings; they do not re-describe the engineering.

Design principle

The professor's paper (Bagler et al., arXiv:2604.28021v1) supplied the *questions and estimators*, not a template of numbers to recover. Work therefore began by reading the paper's Introduction, Figures 1–4, Methods, and Extended Data, then mapping each paper quantity onto CocktailDB columns. Where a variable was absent (cooking techniques, utensils, instruction steps, reliable cuisine labels, NER attributes, per-serving nutrients), the analysis was marked **not testable** rather than approximated with a substitute. Where a variable existed (pipe-delimited ingredients, ratings, votes, prep time, nutrient strings), the paper's conceptual definition was implemented on this file and the empirical pattern was allowed to agree, differ, or be weak.

Project layout and reproducibility

All work lives under `eda_project/`. The original CSV is copied to `data/CocktailDB_raw.csv` and is never overwritten. Scripts write a cleaned table, long ingredient table, wide nutrient table, figures (PNG + vector PDF), CSV summaries, and JSON numerical payloads. The research PDF is assembled last by reading those JSON files, so exponents and sample sizes in the text are generated rather than typed by hand. The full run is `python scripts/run_all.py`, which executes scripts 01–11 in order.

Path	Role
data/	Raw CSV (untouched original copy), CocktailDB_clean.csv, ingredients_long.csv, nutrients_wide.csv
scripts/	Numbered pipeline: audit → clean → EDA → Zipf/Heaps/MA → nutrition → co-occurrence → multivariate → PDF
figures/	All numbered figures as PNG (300 dpi) and PDF (vector)
tables/ and tables/json/	CSV summaries and machine-readable results used by the report builder
report/	CocktailDB_EDA_Research_Report.pdf
README.md, requirements.txt	Reproduction notes and Python dependencies

Pipeline architecture (what each script does)

The pipeline is linear and idempotent: each stage reads the raw or cleaned table and writes artifacts for downstream stages. Shared helpers live in `paths.py`, `plot_style.py` (academic matplotlib defaults), `ingredient_norm.py`, `parsers.py` (votes, prep time, nutrients), and `stats_utils.py`.

Script	What it does
01_data_audit.py	Load raw CSV. Shape, dtypes, unique counts, missingness, exact duplicates, Name vs Name.1 identity, Name+Ingredients clones, same-name variants, rating–vote joint missingness, cuisine/category sparsity, paper-to-column mapping.
02_cleaning.py	Build the analytical table without mutating the raw file. Drop Name.1. Parse votes and prep minutes. Flag Rating=0 & Votes=0 as unrated. Split ingredients; apply four normalization levels. Extract nutrients. Define the frequency corpus.
03_basic_eda.py	Missingness bar and matrix. Recipe-size histogram, violin, ECDF. Top categories (non-missing only). Rating and vote distributions; rating vs log1p(votes). Prep-time distributions and Spearman associations.
04_ingredient_analysis.py	Recipe-incidence frequencies f_i (no double-counting within a recipe). Top-30 bars, frequency histogram, log–log rank–frequency plot.
05_zipf.py	f(r)~r^(−α) with log–log OLS over multiple rank windows; Clauset-style MLE on the frequency vector; likelihood-ratio checks vs log-normal and exponential.
06_heaps.py	100 random recipe orders. Cumulative unique ingredients V(R). Fit β on log–log V vs R (R≥20). Mean curve, 95% permutation band, β histogram.
07_menzerath.py	s_i=−log p(i), L=unique ingredients, y=mean information. Scatter, binned means, nonlinear least squares for y=a L^b exp(cL). Sensitivity across normalization and p(i) definitions.
08_nutrition.py	Parse two nutrient string formats; map Carbs↔Carbohydrates. Descriptives, completeness, calorie/macro plots, Spearman nutrient matrix, KS fits (log-normal, gamma, Weibull, Gaussian, exponential) on x>0.
09_cooccurrence.py	Top-40 co-occurrence heatmap; PMI with a minimum joint-count threshold; network of positive-PMI edges.
10_multivariate.py	Pre-specified Spearman pairs (complexity, popularity, nutrients) with bootstrap CIs. Optional PCA on complete-case standardized nutrients.

Script	What it does
12_extended_additions.py	Prep-time boxplots by top-10 categories; Bayesian vote-adjusted ratings; Zipf log-frequency boxplots by rank stratum.
11_generate_report.py	Compose this PDF (ReportLab): narrative, embedded figures, captions, comparison table. Numbers loaded from JSON.

First inspection (before any modeling)

The raw file has 16,895 rows and 9 columns. Header duplication produced pandas names `Name` and `Name.1`; these two series were compared programmatically and are identical (0 mismatches). Ingredients are 100% present as pipe-delimited free text (quantities still inside the string). Votes are text (comma grouping such as “1,216”). Prep time mixes “3 minutes”, “5 mins”, seconds, hours, days, duplicated phrases, ranges, and unitless numbers. Nutrients use two textual grammars: compact “184 Calories|3g Fat|8g Carbs” and colon-style “Calories: 239 kcal|Carbohydrates: 14 g|…”. The paper PDF was used to lock estimators: f_i = number of recipes containing ingredient i; Zipf via rank–frequency plus Clauset MLE; Heaps with shuffled incremental sampling; Menzerath–Altmann y(L) = a L^b exp(cL); nutrient log-space KS. Cuisine-level laws were abandoned at this stage because Cuisine is 97.7% missing.

Exploratory data analysis protocol

EDA was treated as a sequence of documented questions, each with an explicit observation count n, not as a gallery of charts. Univariate work covered recipe size, labelled categories, ratings, votes, and parsed preparation time. Bivariate work used Spearman rank correlation as the default association measure because votes, times, and nutrients are skewed; Pearson was reported only as a companion on log1p(votes) vs rating. Bootstrap 95% intervals (400 resamples) accompany key Spearman estimates. Heavy-tailed variables were shown on both linear and log (or log1p) axes. ECDFs were used for discrete recipe size. No 3-D plots and no pie charts. matplotlib only, white background, fixed color palette, 300 dpi PNG plus vector PDF. Every figure in the results sections carries: title, axis labels with units, sample size, and a five-part caption (what it shows, key insight, why it matters, statistical evidence, caveat).

EDA block	Operations
Missingness	Percent-missing bars; binary missingness matrix; combination patterns of Category/Nutrients/Ratings/Votes/Prep_Time/Cuisine; joint rating–vote table.
Duplicates	Exact row clones vs Name+Ingredients clones vs same name with different ingredients. Only the first two classes were removed from the frequency corpus.
Recipe structure	raw vs unique ingredient counts; repeats inside a recipe; histogram, violin/box, ECDF of L.
Categorical fields	Category restricted to non-missing rows; long-tail and cumulative share. Cuisine counted and then ruled out for stratified analysis.
Popularity	Rating histogram (bounded 0–5); votes linear + log1p; scatter rating vs log1p(votes); low-vote vs high-vote extremeness.
Prep time	Parsed minutes; raw + log10 histograms; Spearman vs ingredient count and vs votes. Extremes inspected, not auto-deleted.
Ingredients	Incidence ranking; hapax fraction; log–log rank–frequency; co-occurrence and PMI on a top-N set.
Scaling laws	Zipf range sensitivity; Heaps permutations; Menzerath binned NLS + residual/sensitivity checks.
Nutrition	Format harmonization; completeness; macros; Spearman matrix; candidate distribution KS; PCA if complete cases sufficed.

Preprocessing decisions (what we changed, and what we refused to change)

Kept untouched. The original CSV. Original Ratings, Votes, Prep_Time, Nutrients, and Ingredients strings remain on the cleaned table beside derived fields.

Dropped from models only. Redundant `Name.1`. 17 extra exact duplicate rows and 6 extra Name+Ingredients clones, solely so one scraped formula cannot inflate f_i. 1131 names with different ingredient lists were retained as variants.

Ratings / votes. Commas stripped; 0 malformed vote strings. All 116 rows with Rating=0 also had Votes=0; they were recoded as missing in `rating_analyzed` / `votes_analyzed`, not as genuine zero-star scores. No imputation of ratings, votes, categories, cuisine, or free text.

Preparation time. Tokenized into minutes (seconds/60, hours×60, days×1440; ranges averaged; duplicated phrases de-duplicated when the whole string repeated; unitless numbers treated as minutes). Unparsed strings: 0. Zeros and multi-hour/day values were flagged, not deleted.

Ingredients. Split on "|". Four documented normalization levels: *surface* (lowercase/whitespace only); *standard* (strip leading/trailing quantities and units; main analyses; Cointreau→orange liqueur and Kahlua→coffee liqueur only as obvious aliases); *moderate* (also garnish/prep descriptors); *core* (sensitivity only: ice variants and drop "fresh"). Lime, lime juice, and fresh lime juice were **not** collapsed. Frequency is distinct recipes, not mention counts.

Nutrients. Two parsers (colon labels and compact "value + unit + name"). Carbs and Carbohydrates merged only after confirming both are gram masses. Vitamin A remains in IU where reported. Serving size is not assumed. Log transforms and KS tests use strictly positive values; zeros are counted separately and excluded from log-space.

Libraries and statistical tools

The environment is CPython 3.13. Analyses are ordinary scientific Python; no black-box AutoML. ReportLab builds the paginated PDF. Pillow measures figure aspect ratios so embedded plots are not stretched.

Library	Use in this project
pandas / numpy	Tables, missingness, grouping, frequency corpus, I/O of CSV/JSON.
matplotlib	All research figures (histograms, ECDF, log–log plots, heatmaps, PCA scree/projection).
scipy	Spearman/Pearson, skewness, linregress (log–log OLS), curve_fit (Menzerath–Altmann), kstest, candidate distribution MLE.
powerlaw (Alstott)	Clauset-style discrete power-law MLE on ingredient frequencies; likelihood-ratio tests vs log-normal and exponential.
scikit-learn	StandardScaler + PCA on complete-case nutrient rows only (descriptive, not prediction).
networkx	Positive-PMI co-occurrence graph layout.
statsmodels	Available in the environment; core association tests used scipy to keep estimators transparent.
reportlab / Pillow	Academic PDF layout, tables, captions, page numbers; non-distorted figure embedding.
re / unicodedata	Ingredient quantity/unit stripping; vote commas; prep-time tokens; nutrient format detection.

Quality checks after the run: re-execution from the raw CSV; figure files present; JSON values match captions; NaNs excluded from correlations and log transforms; duplicate formulas not allowed to inflate frequencies; log(x) only for x>0; outliers investigated rather than bulk-deleted; no causal language in captions.

Executive Summary

Dataset. CocktailDB contains **16,895 recipe rows** and nine columns. The two name fields are identical. There are **14,916 unique recipe names**. After removing 17 exact duplicate rows and 6 additional Name+Ingredients duplicates, **16,872 recipes** enter frequency analyses. Same-name recipes with different ingredient lists (1131 names) were retained as variants. Major missingness: Category 59.0%, Nutrients 66.3%, Ratings 52.5%, Votes 52.6%, Prep_Time 58.6%, Cuisine 97.7%. Ingredients are complete.

1. Ingredient usage is concentrated: the most common normalized ingredient is “vodka” (2,191 recipes; 13.0% of the corpus). 73.5% of the vocabulary occurs in only one recipe.

2. Rank–frequency curves are heavy-tailed. A log–log OLS Zipf exponent of $\alpha \approx 1.00$ was obtained on ranks 5–500 (descriptive $R^2 = 0.991$). A Clauset-style power-law fit to the frequency distribution is reported separately and is not treated as automatic confirmation of Zipf’s law.

3. Heaps’ law is clearly sublinear: mean $\beta = 0.738$ (100 permutations; 95% interval [0.725, 0.751]). This is close to the paper’s global $\beta \approx 0.56$ but is estimated on a smaller, cocktail-only corpus and is not a target to match.

4. Menzerath–Altmann: Spearman correlation between recipe size L and mean ingredient information y(L) is $\rho = 0.101$. The paper’s decreasing-then-rising global pattern is not assumed; the fitted MA parameters and R^2 are reported in §6.3.

5. Recipe size is small relative to the paper’s food corpus: mean unique ingredients = 4.67 (median 4.0; n = 16,872).

6. Ratings, where observed, are high and left-censored toward 5 stars (mean 4.32, median 4.45, n = 7,890). All 116 Rating=0 rows have Votes=0 and were treated as unrated.

7. Votes are heavy-tailed. Spearman association of votes with rating is $\rho = -0.058$ (n = 7878). This is an association, not evidence that votes cause ratings.

8. Cuisine cannot support cuisine-level laws: only 393 non-missing values (2.33% of the frequency corpus), with labels that mix drink types and nationalities.

9. Category is available for 6,925 recipes (41.0% of the frequency corpus); category comparisons are conditional on that subset.

10. Nutrient strings exist for 5,694 recipes (66.3% missing). Log-space KS comparisons are reported per nutrient; log-normality is not claimed globally. Best KS candidates include: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm.

11. Strongest complete-case Spearman associations among the pre-specified multivariate pairs: ingredient count vs calories (ρ=0.25, n=5683), ingredient count vs votes (ρ=-0.19, n=7878), ingredient count vs protein (ρ=0.14, n=4988).

12. Cooking techniques, utensils, instruction steps, and 26-cuisine stratification from the reference paper cannot be reproduced because those variables are absent.

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Abstract

Bagler and colleagues argue that traditional recipes behave as a compositional symbolic system: ingredient frequencies are Zipf-like, culinary vocabulary grows sublinearly (Heaps' law), recipe size trades off with average ingredient information (Menzerath–Altmann), and macronutrient amounts are approximately log-normal. This study asks whether those regularities appear in CocktailDB, a corpus of 16,895 cocktail-related recipe rows with pipe-delimited ingredients and partially observed ratings, votes, preparation times, categories, cuisines, and nutrient strings. Methods were adapted, not copied: frequency is the number of distinct recipes containing a normalized ingredient; Heaps curves use 100 random recipe orders; Menzerath–Altmann uses y(L)=mean[−log p(i)]; nutrient log-normality is tested only on positive extracted values. Technique- and cuisine-stratified analyses from the paper are not testable here. Cocktail recipes are short (median 4 unique ingredients). Ingredient frequencies are heavy-tailed (OLS Zipf $\alpha \approx 1.00$ on a restricted rank range). Vocabulary growth is sublinear ($\beta \approx 0.74$). The Menzerath–Altmann relationship is estimated rather than assumed to match the global food corpus. Nutrient coverage is incomplete and two text formats had to be harmonized. The contribution is an honest mapping from a global culinary-law framework onto a cocktail dataset, including negative and non-testable results.

1. Introduction

The reference paper treats recipes as symbolic compositions analogous to sentences: discrete culinary units are combined under cultural and practical constraints. In a corpus of 118,083 traditional recipes from 26 cuisines, the authors report Zipf-like ingredient usage (global rank–frequency exponent $\alpha \approx 1.53$; cuisine-mean ≈ 1.39), Heaps' law with $\beta \approx 0.56$ (cuisine-mean ≈ 0.54), a Menzerath–Altmann relation y(L)=a L^b exp(cL) with a global minimum near $L \approx 13$, and log-normal macronutrient concentrations. Those claims concern a NER-structured, cuisine-labelled food corpus, not cocktails.

Cocktails are a narrower compositional domain: typically few ingredients, repeated base spirits, juices, liqueurs, and garnishes, and recipe sources that often mix blogs, databases, and user reviews. If Zipf/Heaps structure is truly generic to culinary design, a cocktail corpus should show heavy-tailed reuse and sublinear vocabulary growth even without global cuisine coverage. If instead those laws depend on savory cooking, long ingredient lists, or NER-normalized taxonomies, cocktail data may differ—and that difference is scientifically informative.

What can and cannot be reproduced. Reproducible in spirit: ingredient frequency, Zipf rank–frequency analysis, Heaps growth, Menzerath–Altmann on recipe size versus mean ingredient information, nutrient distributional tests where values exist, and recipe-size histograms. Not reproducible: named-entity decomposition of quantity/unit/state; cooking-technique and utensil vocabularies; instruction-step complexity; geographic maps; 26-cuisine exponent distributions; per-serving RecipeDB nutrient standardization; and the paper's generative simulations as a reproduction. Cuisine in CocktailDB is 97.7% missing and is not a substitute for the paper's cuisine axis.

2. Research Objectives

The objectives are: (i) audit CocktailDB and map paper variables onto observed columns; (ii) quantify missingness without imputing ratings, votes, or text fields; (iii) standardize ingredients with a documented hierarchy and without collapsing lime / lime juice / fresh lime juice; (iv) describe recipe size, category, rating, vote, and preparation-time distributions on observed subsets; (v) test Zipf, Heaps, and Menzerath–Altmann structure; (vi) parse nutrients, assess completeness, and compare candidate distributions; (vii) add co-occurrence and multivariate association analyses that the paper does not emphasize; (viii) state where the cocktail corpus agrees, differs, or cannot be compared.

3. Dataset Description

The file contains 16,895 rows and 9 columns: Name, Name.1, Category, Ingredients, Nutrients, Ratings, Votes, Prep_Time, Cuisine. Name equals Name.1 on every row (mismatches = 0). Name.1 is redundant and is dropped from models. Ingredients are pipe-delimited free-text phrases with quantities and units still inside the string (unlike the paper's NER attributes). There is no identifier column beyond the recipe name. Votes are stored as text (including comma grouping). Ratings are numeric on a 0–5 scale.

Paper quantity	CocktailDB field	Status
Recipe document	Row; Name	Present
Ingredient identity (NER)	Derived from Ingredients	Approximate; not NER
Frequency f_i	Distinct recipes containing i	Implemented
Recipe size L	unique_ingredient_count	Implemented
Cuisine (26 classes)	Cuisine	Present but 97.7% missing
Cooking techniques	—	Absent

Paper quantity	CocktailDB field	Status
Utensils / steps	—	Absent
Macronutrients per serving	Nutrients text	Partial; serving not documented
Cook time	Prep_Time	Partial; heterogeneous units

4. Data Quality and Preprocessing

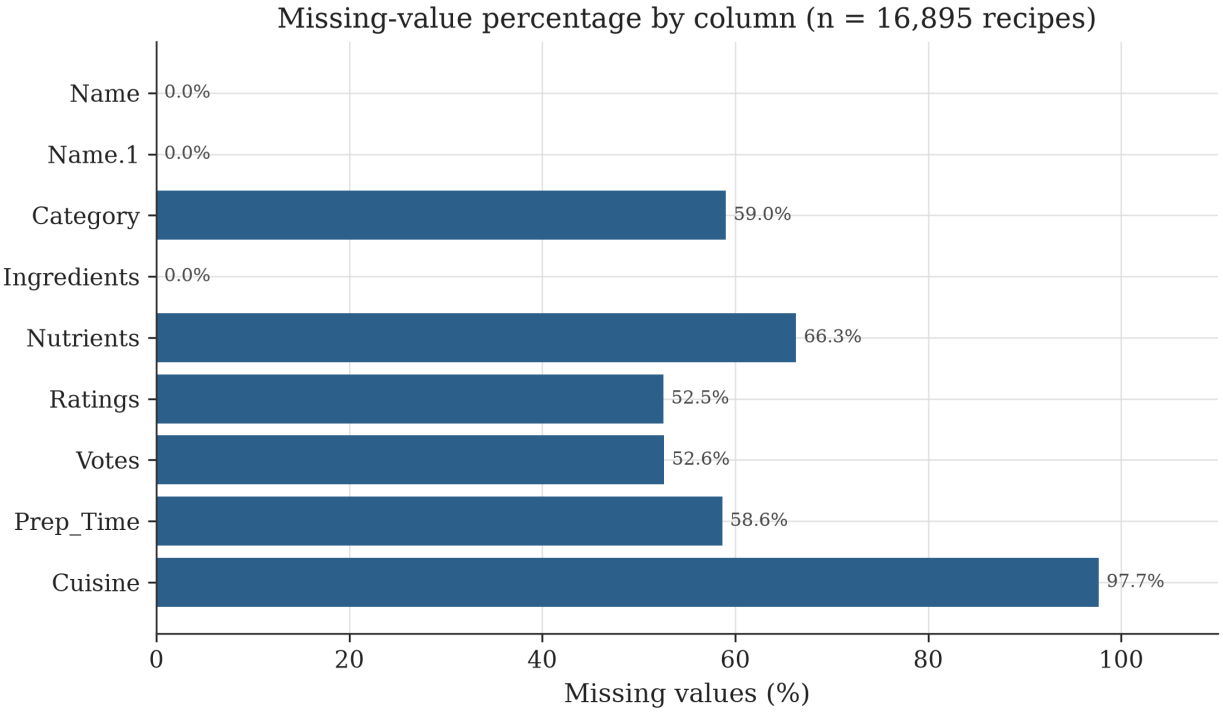
4.1 Audit of duplicates and redundancy

Exact duplicate rows: 17 extras (33 rows involved). These are scrape/storage repeats and were dropped from the frequency corpus. Name+Ingredients duplicates beyond the first: 23. These are the same formula under the same title and were also dropped for frequency counts so that one recipe cannot inflate f_i. Duplicate names overall: 1979 extra rows; 1144 names appear more than once. Of those, 1131 names have different ingredient strings (legitimate variants, retained) and 13 names repeat the same ingredients. Name and Name.1 are identical; Name.1 is excluded from analysis.

4.2 Missingness

Ratings and votes are almost a joint field: both present in 8,005 rows, both missing in 8,878, ratings without votes in 12, votes without ratings in 0. This is systematic missingness of a popularity block, not independent gaps. Cuisine is too sparse for stratified laws (393 non-missing). Category missingness (59.0%) means category plots are not globally representative.

Figure 1. Missing-value percentage by column.



What this figure shows. *Each bar is the percentage of the 16,895 rows with a null in that column. Ingredients and both name fields are complete; Cuisine is almost empty.*

Key insight. *Popularity (ratings/votes), category, prep time, and nutrients are majority-missing. Cuisine missingness is 97.7%.*

Why it matters. *Any comparison that ignores missingness would pretend CocktailDB is a complete design matrix comparable to the paper's curated corpus.*

Statistical evidence. *n = 16,895 rows. Highest missingness: Cuisine 97.7%, Nutrients 66.3%, Category 59.0%, Prep_Time 58.6%, Ratings 52.5%.*

Caveat. *Null encoding is taken as missing; empty strings were not separately treated as a second missing class for these fields.*

Figure 2. Missingness pattern (downsampled rows).



What this figure shows. *Black cells are missing. Rows are shown at a regular stride to make blocks visible. Ingredients stay complete while Cuisine is almost continuously missing.*

Key insight. *Missingness is structured: large contiguous stretches lack category/nutrients/ratings together, consistent with heterogeneous source tables concatenated into one file.*

Why it matters. *Systematic missingness implies that nutrient and rating analyses describe source subsets, not a random sample of all cocktail recipes.*

Statistical evidence. *Displayed stride chosen so that ~900 row-samples represent 16,895 recipes. Joint rating–vote presence n = 8,005.*

Caveat. *Downsampling can hide single-row anomalies; combination counts in the appendix use the full matrix.*

4.3 Ratings, votes, and preparation time

All 116 rows with Rating = 0 also have Votes = 0. They are coded as unrated (rating_analyzed and votes_analyzed missing), not as genuine zero-star quality. The original columns are preserved. Votes such as “1,216” were parsed by removing commas; malformed votes = 0. Prep times were parsed into minutes (seconds/60, hours×60, days×1440; ranges averaged; unitless numbers treated as minutes). On the frequency corpus, prep-time minutes have min 0.00, median 3.00, mean 7.95, IQR 0.00, P90 5.00, P95 10, max 20,170 (n = 6976). Extremes include multi-hour and multi-day values and one 336-hour string; they were retained and flagged rather than deleted. Unparsed prep strings: 0. Zeros: 44.

4.4 Ingredient normalization

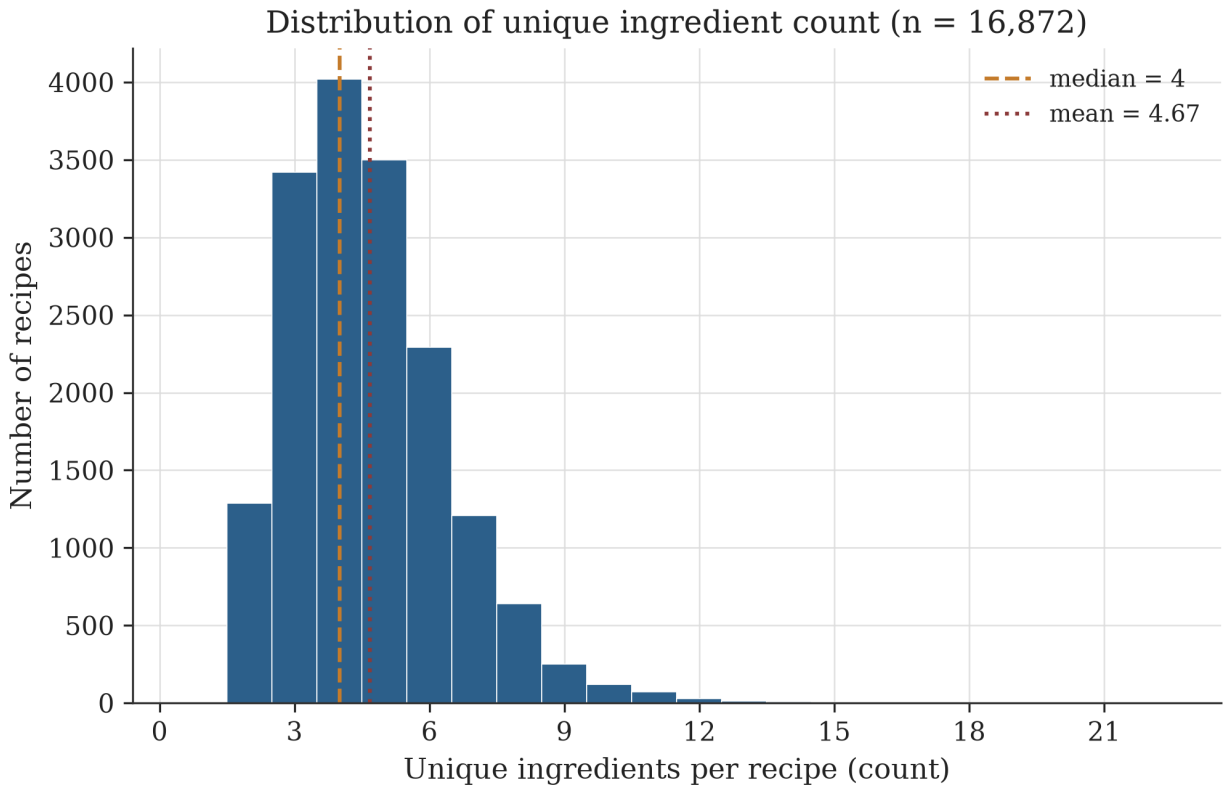
Four levels were computed. Main analyses use **standard** (lowercase, whitespace, strip leading/trailing quantities and units; Cointreau→orange liqueur and Kahlua→coffee liqueur only as obvious aliases). Moderate additionally strips garnish/prep descriptors. Core is sensitivity-only (ice variants; drop “fresh”). Vocabulary at standard level: 14,466 unique strings from 79,411 mentions. Surface unique strings: 31,180. Core unique: 12,480. Lime versus lime juice is not collapsed.

5. Exploratory Data Analysis

5.1 Recipe structure

Observation: cocktail formulas are short. Mean unique ingredients = 4.67, median = 4.00, IQR = 3.00, range 1–22 (n = 16,872). The paper’s global food corpus peaked near 8–12 ingredients with mean about 10; cocktails sit well below that characteristic scale. Interpretation: drink recipes are composed around a small number of components. Limitation: unique_ingredient_count depends on normalization; residual quantity fragments would bias counts upward.

Figure 3. Distribution of unique ingredients per recipe.



What this figure shows. *Histogram of L on the frequency corpus (n = 16,872). A vertical median and mean are overlaid.*

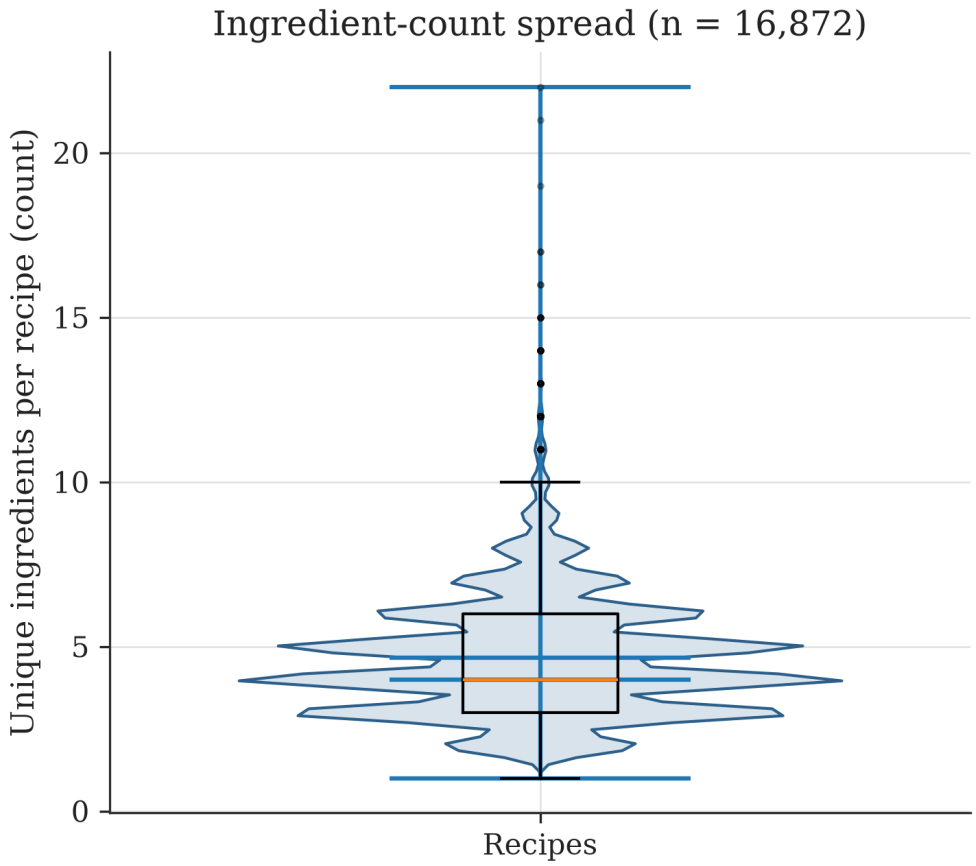
Key insight. *Most recipes use about 4 unique ingredients; the right tail reaches 22.*

Why it matters. *Recipe complexity in this corpus is primarily low-dimensional, which will later constrain Menzerath–Altmann (few large-L bins).*

Statistical evidence. *n = 16,872; mean = 4.67; median = 4.00; SD = 1.80.*

Caveat. *L is unique normalized ingredients, not raw pipe tokens.*

Figure 4. Spread of recipe size.



What this figure shows. *Violin plus boxplot of unique ingredient counts.*

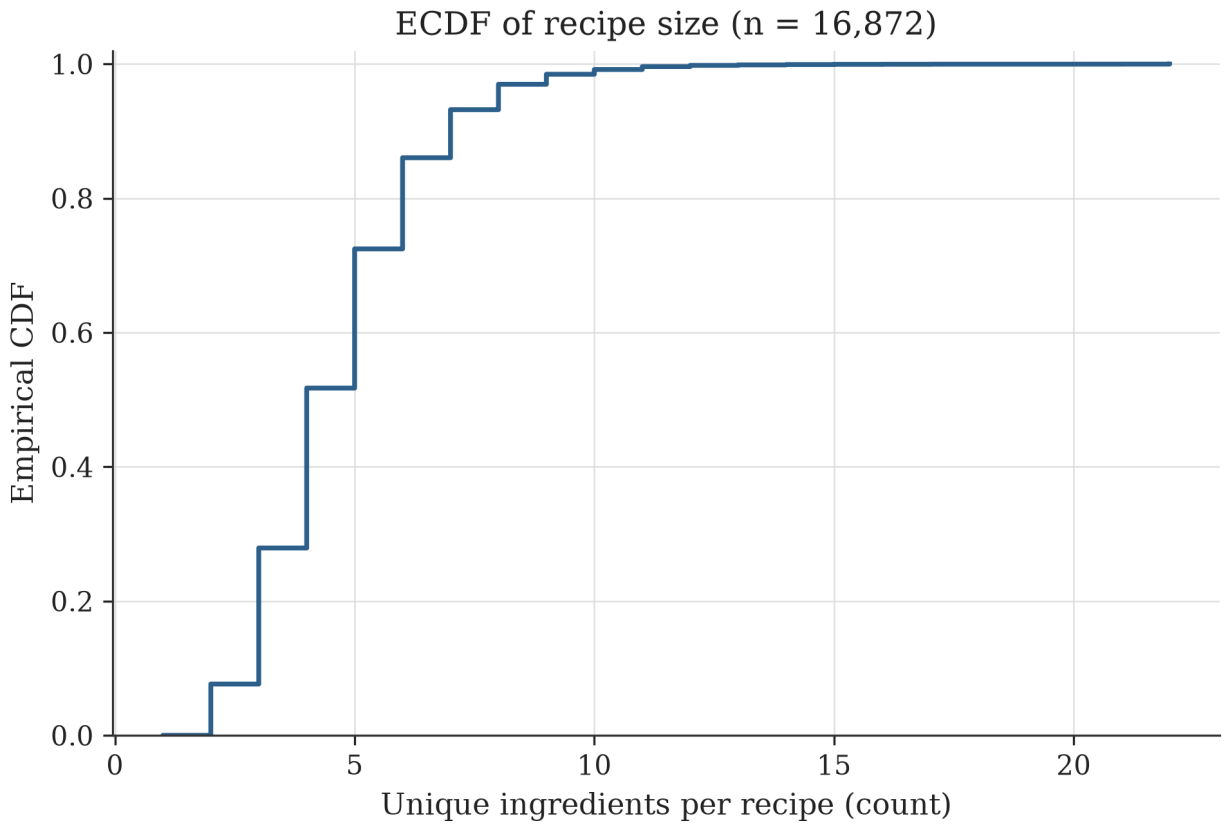
Key insight. *Dispersion is modest (IQR 3.00); outliers exist but do not dominate the mass.*

Why it matters. *Outlier long recipes exist (punches, pitchers, garnished showpieces) but are uncommon.*

Statistical evidence. *n = 16,872; min = 1; max = 22.*

Caveat. *Boxplot whiskers follow the default matplotlib convention; outliers are plotted, not removed.*

Figure 5. ECDF of recipe size.



What this figure shows. *The empirical CDF of L shows how quickly probability mass accumulates at small sizes.*

Key insight. *A large fraction of recipes have L at or below the median of 4.*

Why it matters. *ECDFs avoid binning artifacts in a discrete, skewed count variable.*

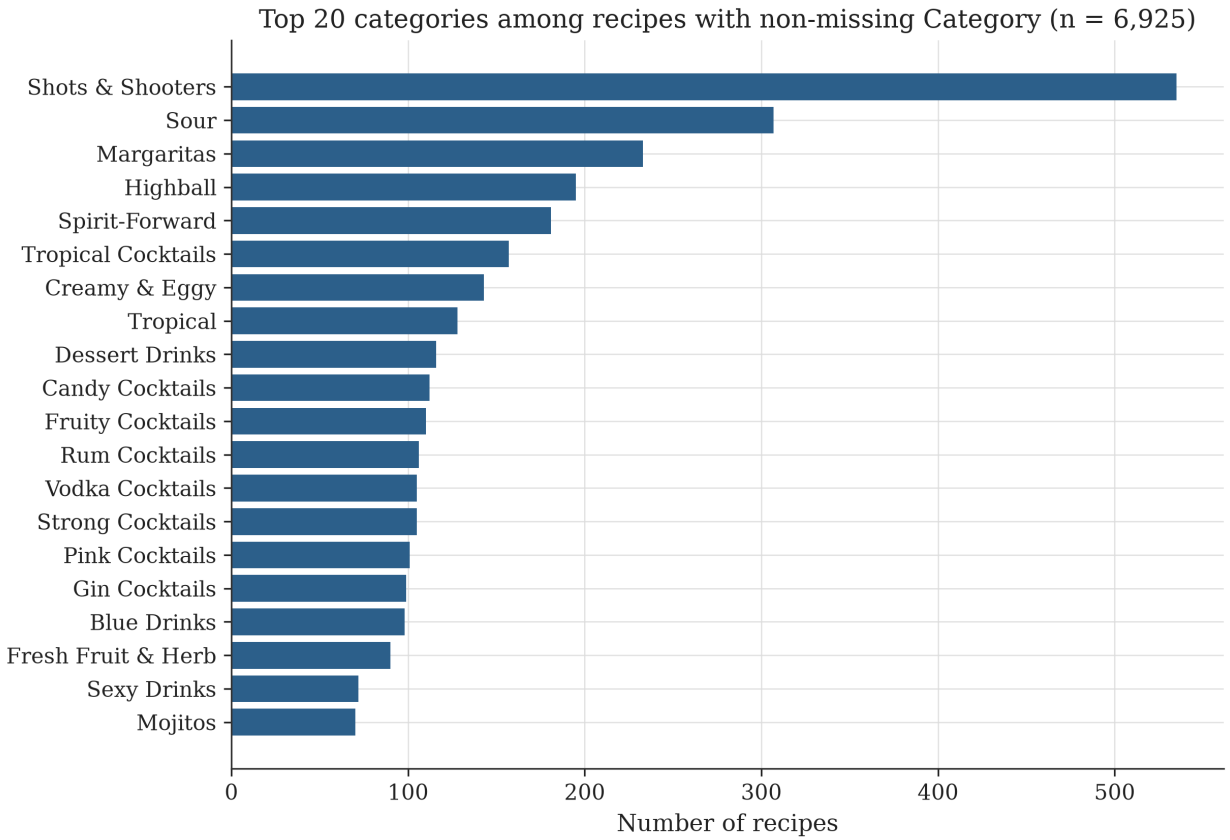
Statistical evidence. *n = 16,872.*

Caveat. *Ties at integer L produce vertical jumps.*

5.2 Categories

Category is observed for 6,925 recipes (41.0% of the frequency corpus), with 429 distinct labels. The top five categories account for 21.0% of labelled recipes; the top 20 account for 44.2%. Labels mix glassware, flavor families, and spirit families (shots, sours, margaritas, highballs). Missing Category is not treated as a category. Conclusions are conditional on the labelled subset.

Figure 6. Top 20 categories by recipe count.



What this figure shows. *Counts among recipes with non-missing Category (n = 6,925).*

Key insight. *The leading labelled class is dominated by a long-tailed remainder: 429 labels overall.*

Why it matters. *Category is a noisy ontology, useful for describing the labelled slice, not for claiming a global taxonomy of cocktails.*

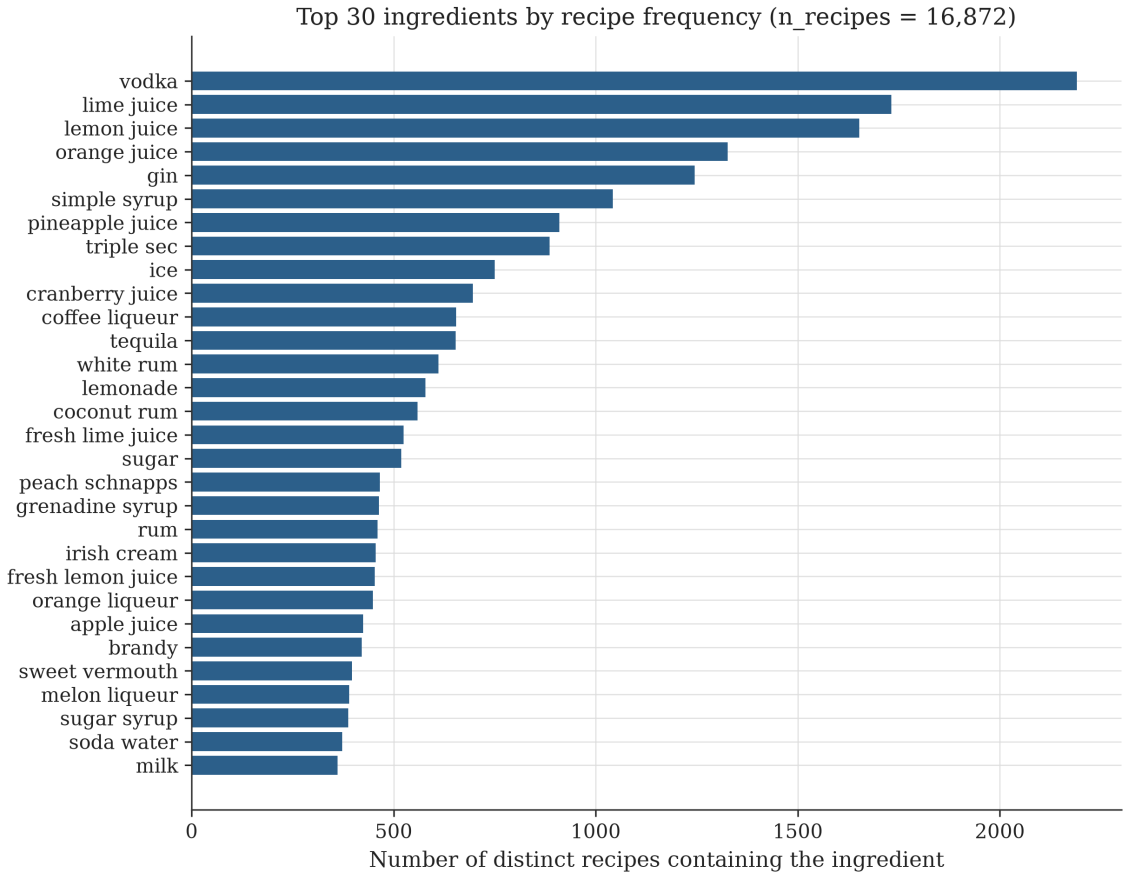
Statistical evidence. *n_labelled = 6,925; coverage = 41.0% of 16,872 frequency-corpus recipes; top-20 share = 44.2%.*

Caveat. *Do not compare categories as if every recipe were labelled.*

5.3 Ingredients

Using standard normalization, 14,466 distinct ingredients appear in 16,872 recipes. “vodka” occurs in 2,191 recipes (13.0%). Hapax ingredients (frequency 1) are 10,626 (73.5% of vocabulary). This is the same conceptual frequency as the paper: f_i = number of distinct recipes containing i, with within-recipe repeats not double-counted.

Figure 13. Top 30 ingredients by recipe frequency.



What this figure shows. *Bars are f_i for the 30 most common standard-normalized ingredients (n_recipes = 16,872).*

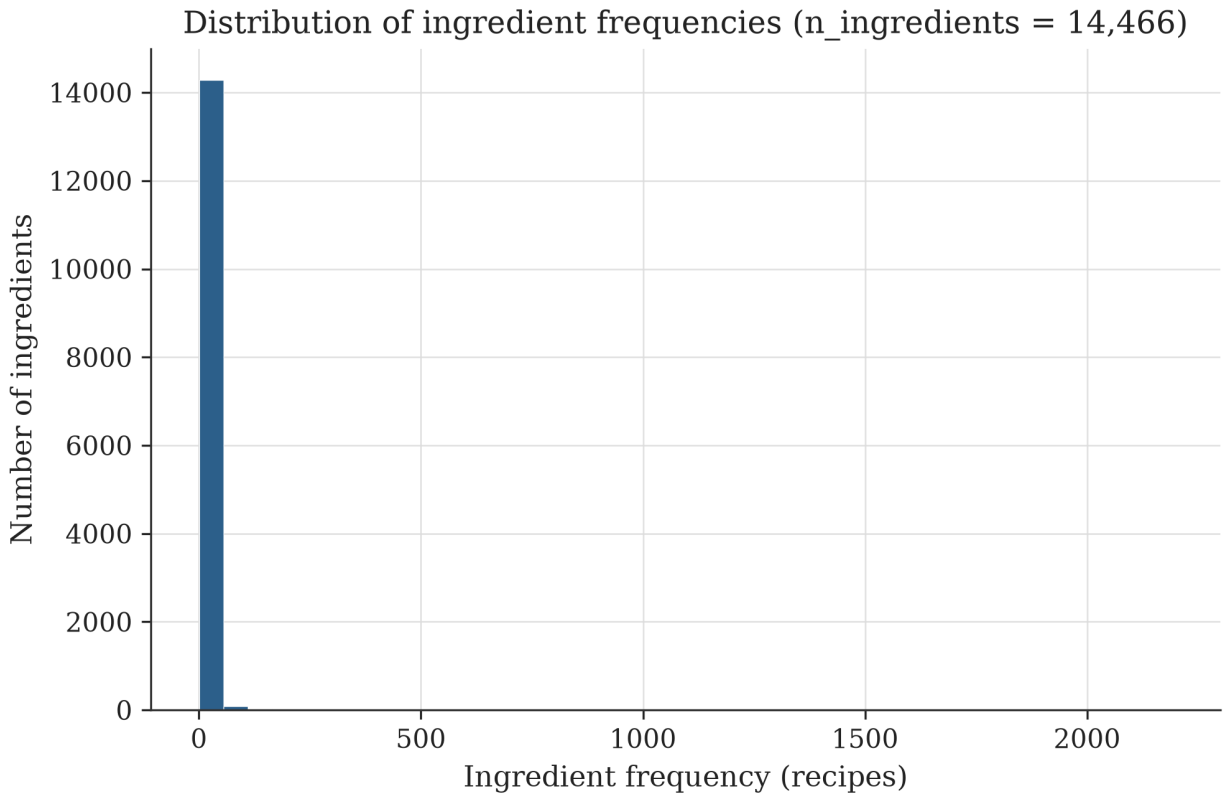
Key insight. *A small set of bases, juices, sweeteners, bitters, and ice-like items dominate. The leader is vodka with f = 2,191.*

Why it matters. *Preferential reuse of a core bar vocabulary is the cocktail analogue of the paper's salt/onion/butter/flour/oil dominance in food.*

Statistical evidence. *n_recipes = 16,872; vocabulary = 14,466; top-10 share of all frequency mass = 15.8%.*

Caveat. *Normalization aliases (e.g., Cointreau as orange liqueur) slightly inflate some totals; lime juice is kept distinct from lime.*

Figure 14. Distribution of ingredient frequencies.



What this figure shows. *Most ingredients are rare; a thin tail of high-frequency items sits far to the right.*

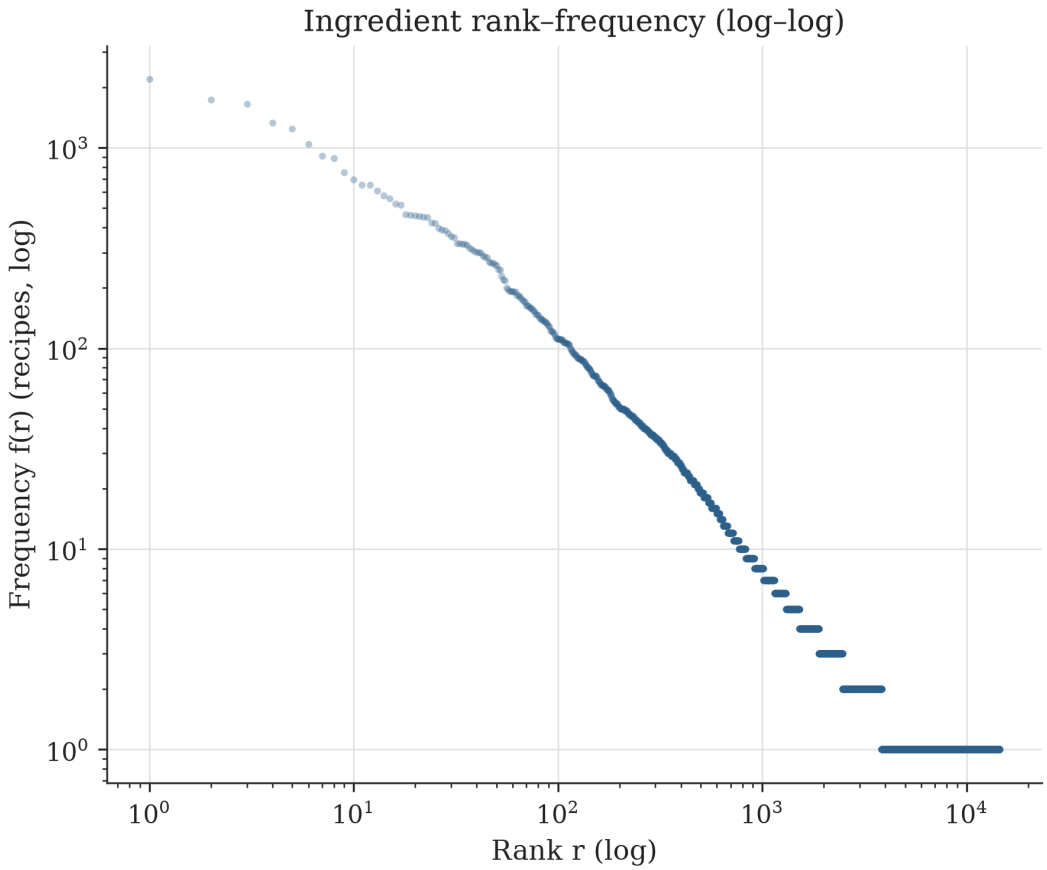
Key insight. *73.5% of ingredients appear once. Mean frequency = 5.45; median = 1.00.*

Why it matters. *This is the size distribution that Zipf and Heaps later summarize: reuse of few items plus a long tail of novelty.*

Statistical evidence. *n_ingredients = 14,466.*

Caveat. *The linear-axis histogram compresses the tail; the rank–frequency plot is the appropriate scaling view.*

Figure 15. Rank–frequency plot on log–log axes.



What this figure shows. *Each point is an ingredient: rank r versus recipe frequency f(r).*

Key insight. *The cloud declines over several decades and is curved rather than a single perfect line, which already warns against an unrestricted power-law claim.*

Why it matters. *Heavy tails mean a bartender's core shelf covers many recipes while rare modifiers create diversity.*

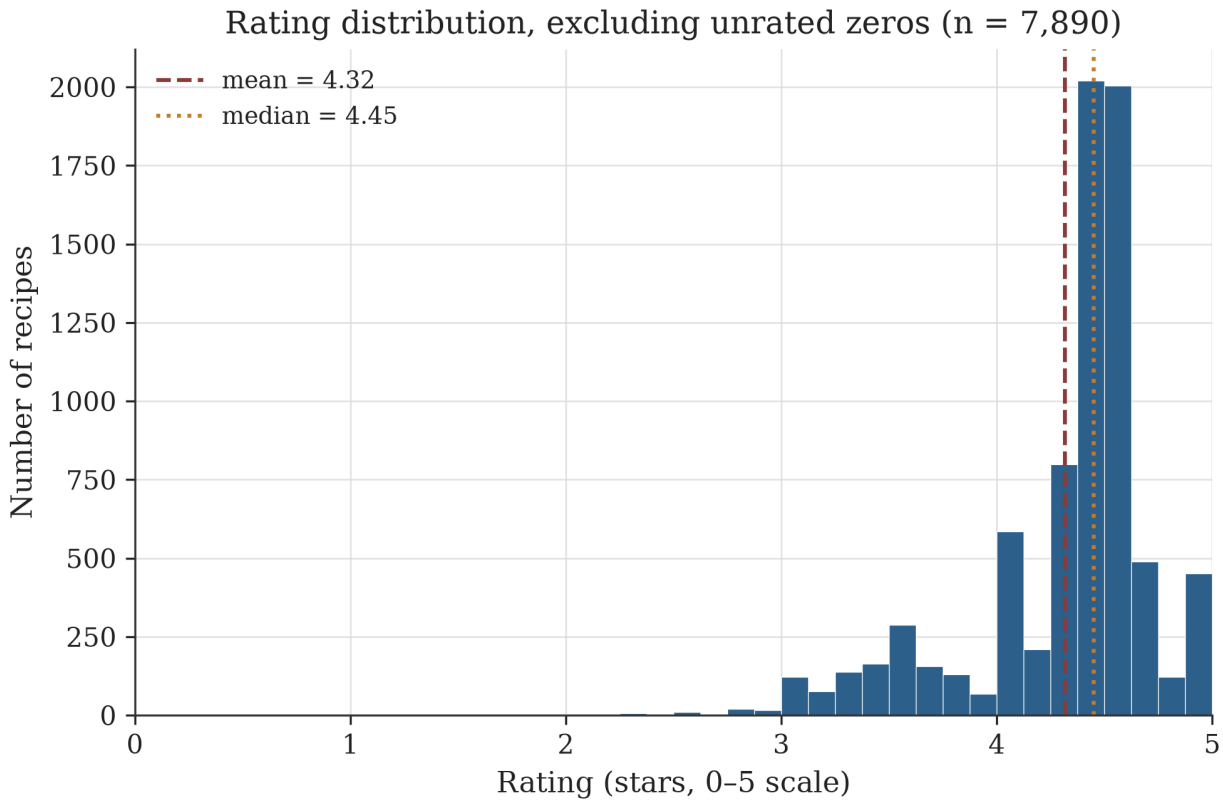
Statistical evidence. *n = 14,466 ingredients; n_recipes = 16,872.*

Caveat. *Visual linearity is not a statistical test; see Figure 16.*

5.4 Ratings and popularity

Ratings (excluding unrated zeros): n = 7,890, mean = 4.317, median = 4.450, SD = 0.442, IQR = 0.320, skewness = -1.441. The support is bounded by 5, so unbounded continuous distribution fitting is not used as a primary model. Votes: n = 7,878, median = 30, mean = 38.4, max = 4,348, skewness = 25.10. Pearson(rating, log1p votes) r = -0.060, p = < 0.0001, n = 7878. Spearman(rating, votes) ρ = -0.058, p = < 0.0001, n = 7878; bootstrap 95% CI [-0.08157368286926583, -0.030059730233818565]. Spearman is primary because votes are heavy-tailed. Low-vote recipes (≤3 votes, n = 540) have more extreme ratings (share of ratings ≤2 or ≥4.9 = 60.4%) than high-vote recipes (≥50, n = 1325, share 0.5%).

Figure 7. Rating distribution.



What this figure shows. *Histogram of analyzed ratings on 0–5 (n = 7,890), excluding Rating=Votes=0.*

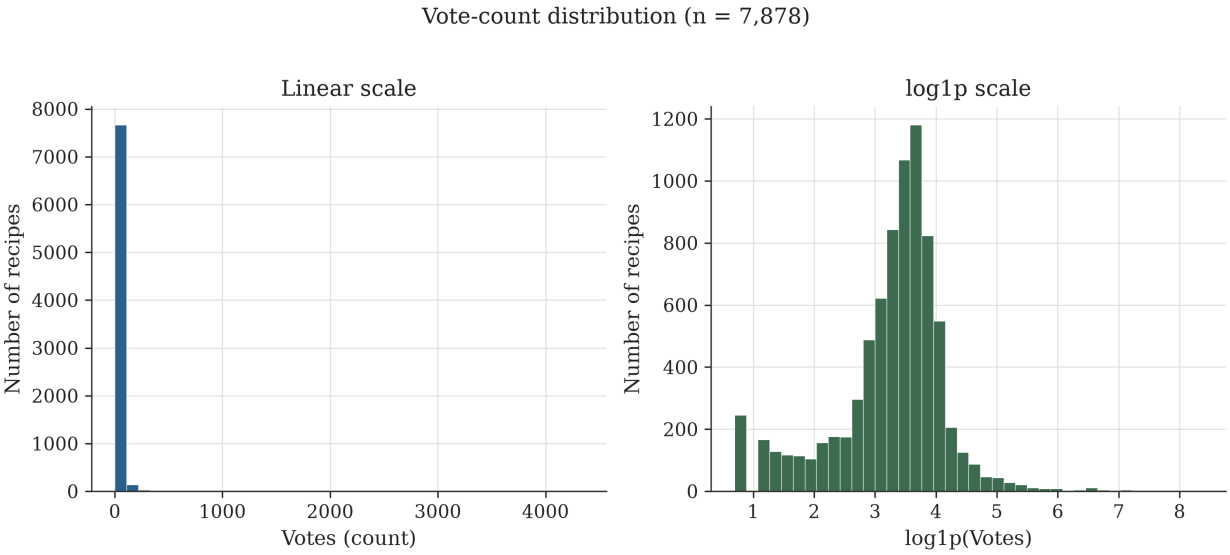
Key insight. *Mass piles near 4–5 stars (median 4.45). The scale is compressed at the top.*

Why it matters. *High means do not imply uniformly excellent recipes; they can reflect selection, interface defaults, and small-n 5-star ratings.*

Statistical evidence. *mean = 4.317; SD = 0.442; IQR = 0.320; skewness = -1.441.*

Caveat. *Bounded discrete/ordinal ratings; unrated zeros excluded by design.*

Figure 8. Votes on linear and log1p scales.



What this figure shows. *Left: raw vote counts. Right: log1p(Votes). n = 7,878.*

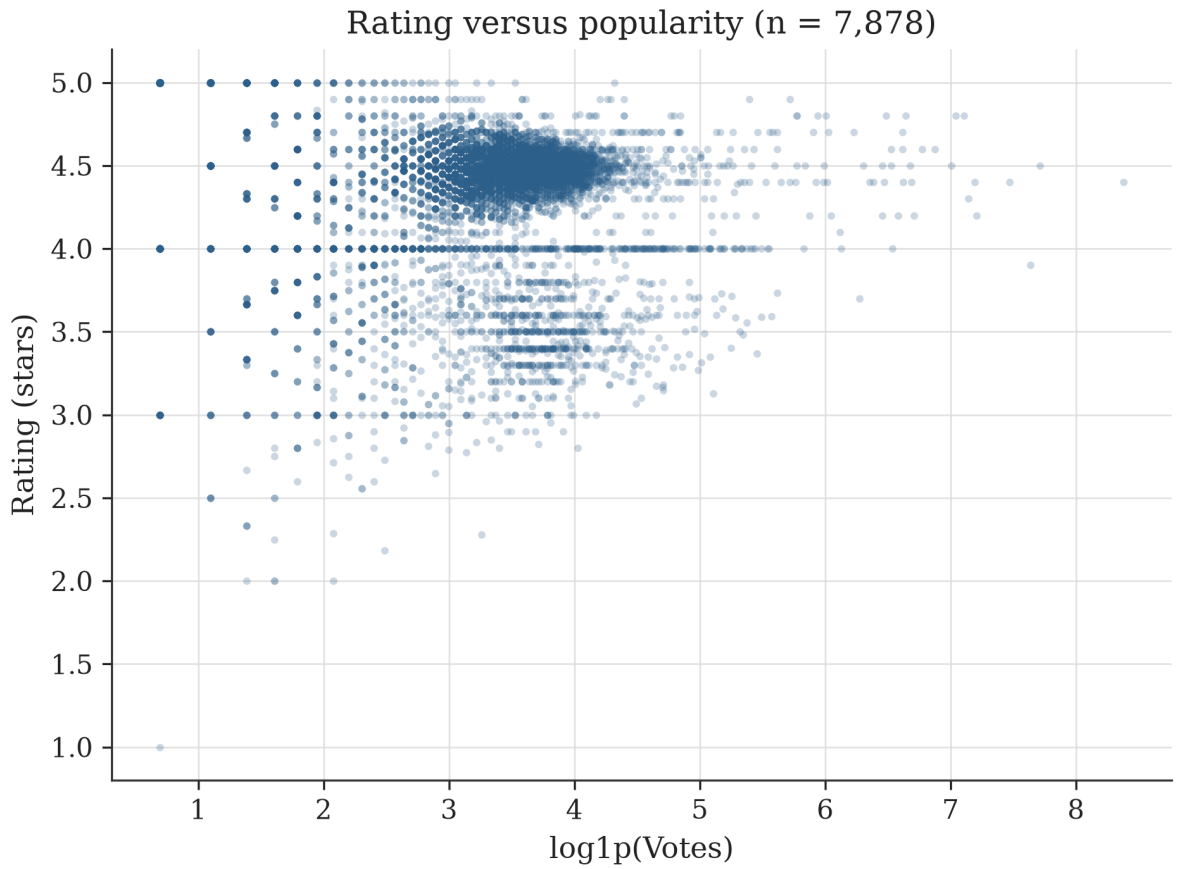
Key insight. *Popularity is long-tailed (mean 38.4 ■ median 30; max 4,348).*

Why it matters. *A few recipes attract most attention; vote counts are a poor proxy for objective quality.*

Statistical evidence. *skewness = 25.10; P90 = 58.*

Caveat. *Votes exist only for the popularity-observed subset.*

Figure 9. Rating versus log1p(Votes).



What this figure shows. *Each point is a recipe with both an analyzed rating and vote count.*

Key insight. *Association is positive but modest (Spearman ρ = -0.058). Low-n ratings fan out to 1 and 5.*

Why it matters. *Highly voted recipes are not guaranteed higher ratings; they do tend to occupy a narrower high-rating band, consistent with averaging.*

Statistical evidence. *Spearman ρ = -0.058, p = < 0.0001, n = 7878; Pearson(r, log1p votes) = -0.060.*

Caveat. *No causal claim: better-known recipes may be better, or only more rated.*

Vote-adjusted ratings

Following the extended recipes EDA notebook, a Bayesian-style shrinkage estimator is adj = (V·R + m·C) / (V + m) with m = 10 and C = global mean analyzed rating = 4.316 (n = 7,878 recipes with both rating and votes after unrated-zero exclusion). This does not create new information; it down-weights 1-vote 5-star recipes relative to well-voted ones.

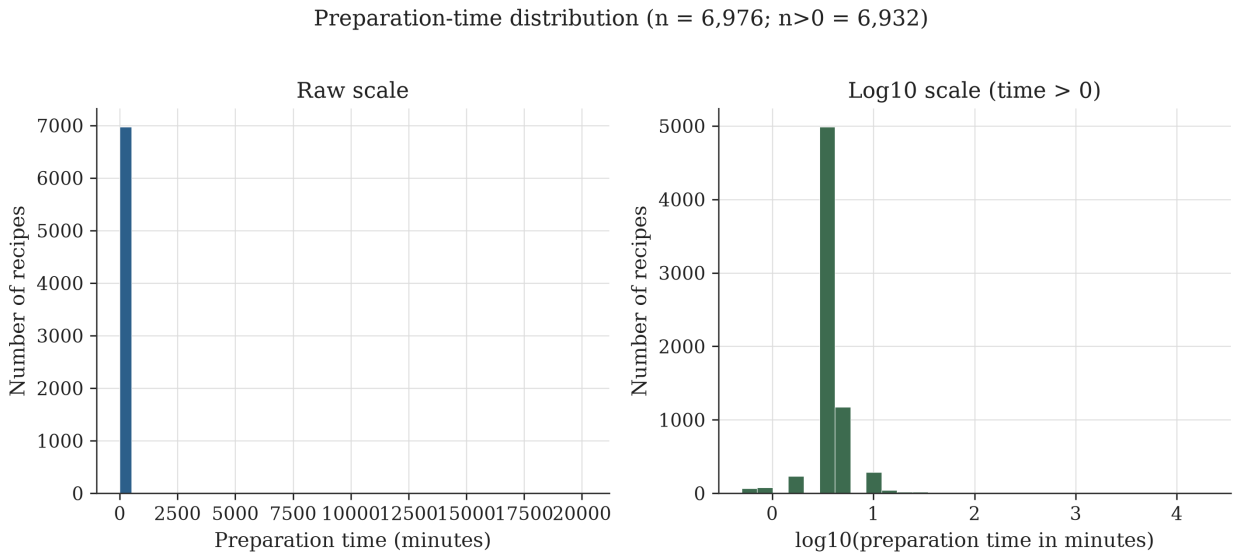
Name	Rating	Votes	Adj. rating
El Diablo Cocktail	5.00	74	4.919
Simple Syrup	4.90	304	4.881
Grandma's Apple Pie 'Ala Mode' Moonshine	4.90	219	4.875
Skeleton Key Cocktail	5.00	33	4.841
Whiskey Slush	4.90	80	4.835
Easy Frozen Margaritas	4.90	72	4.829
Sex on the Beach	4.90	70	4.827
Irish Old Fashioned	4.90	66	4.823
Vruca Rakija Recipe	5.00	27	4.815
Easy Mojitos	5.00	24	4.799

Table. Top 10 recipes by vote-adjusted rating (m = 10).

5.5 Preparation time

Parsed prep time (minutes) on the frequency corpus where available: n = 6,976, median = 3.00, mean = 7.95, max = 20,170. Spearman(prep time, ingredient count) ρ = 0.352, p = < 0.0001, n = 6976. Spearman(prep time, votes) ρ = -0.290, p = < 0.0001, n = 6082. Neither association is interpreted as a production function or a popularity mechanism.

Figure 10. Preparation-time distribution.



What this figure shows. *Raw minutes (left) and log10 of positive times (right). n = 6,976.*

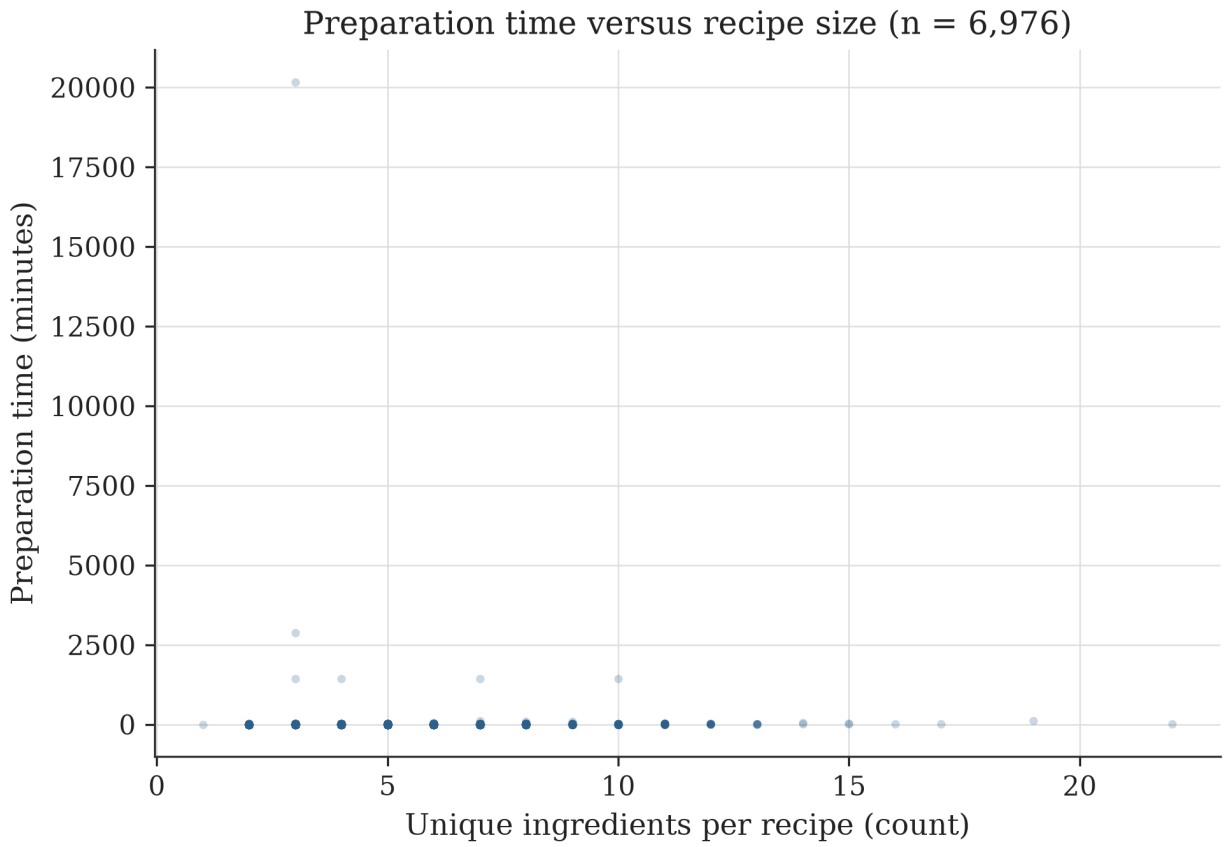
Key insight. *Typical times are a few minutes (median 3.0), with a long right tail including hours/days.*

Why it matters. *Most cocktails are quick to assemble; extreme values may mix infusion/rest times with active prep or parsing artifacts (e.g., duplicated phrases, 336 hours).*

Statistical evidence. *mean = 7.95; P95 = 10; max = 20,170.*

Caveat. *Unitless numbers were assumed to be minutes; that assumption is documented, not verified by the source.*

Figure 11. Preparation time versus ingredient count.



What this figure shows. *Scatter of minutes against unique ingredient count.*

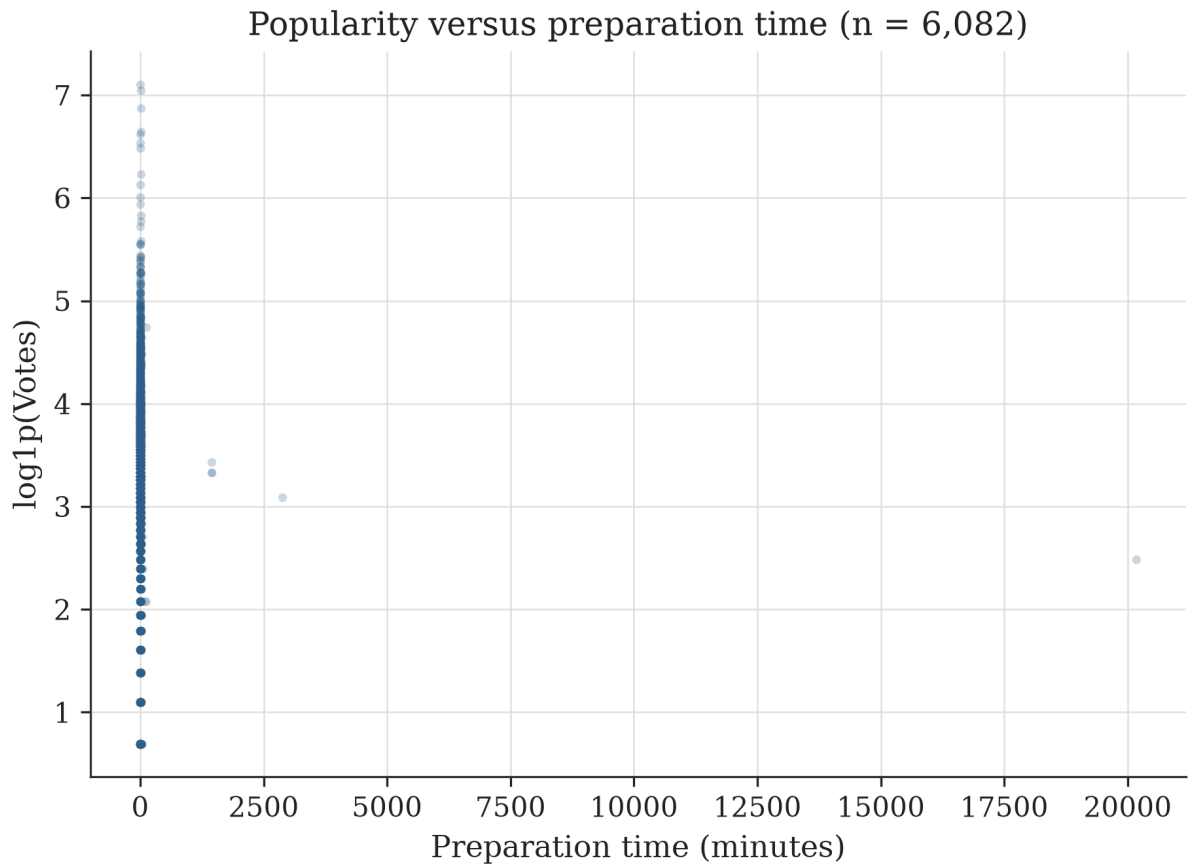
Key insight. *The rank correlation is 0.352 (n = 6976).*

Why it matters. *A weak relationship would mean longer drinks are not simply those with more ingredients (batching, chilling, or infusion can dominate).*

Statistical evidence. *Spearman ρ = 0.352, p = < 0.0001; 95% bootstrap CI [0.3324295891922727, 0.37284437718635].*

Caveat. *Outliers in time can dominate Pearson correlation; Spearman is used.*

Figure 12. Popularity versus preparation time.



What this figure shows. *log1p(Votes) against parsed minutes.*

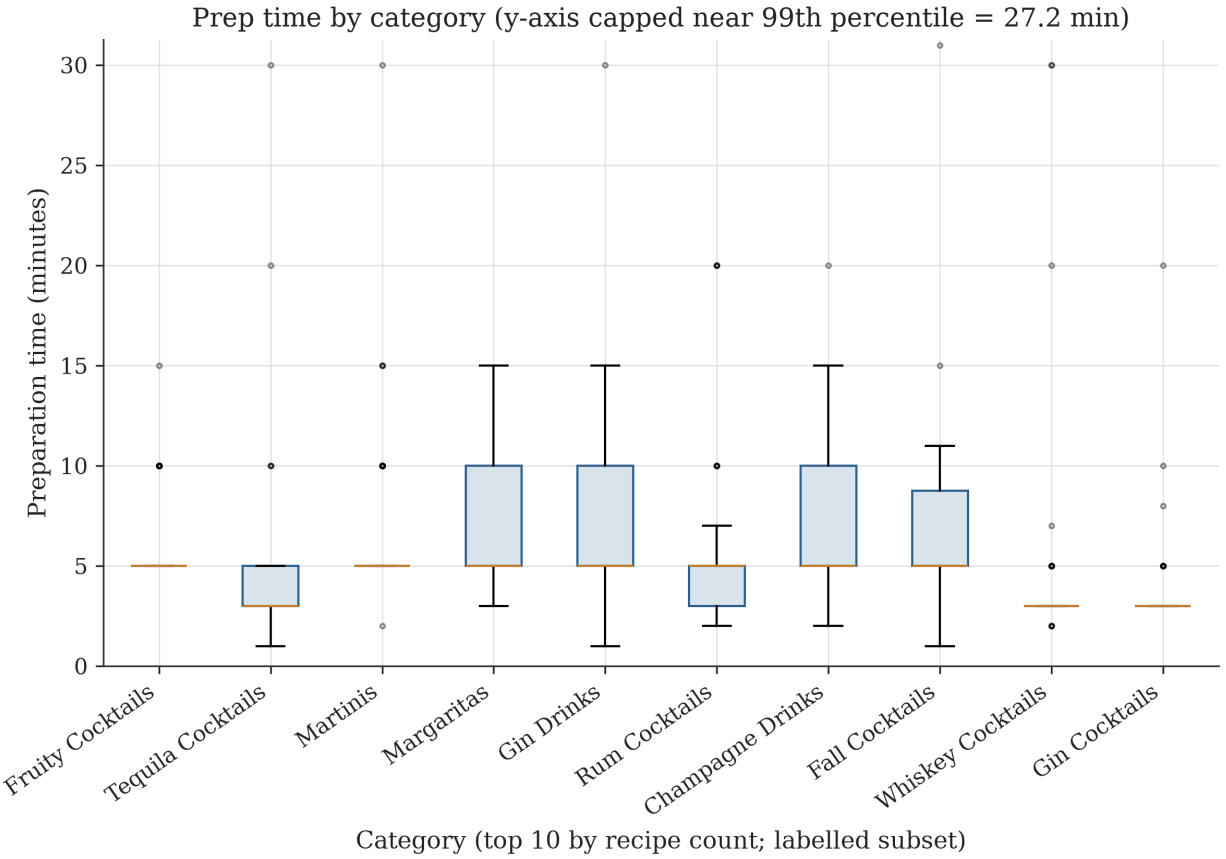
Key insight. *Spearman ρ = -0.290 (n = 6082).*

Why it matters. *Even a statistically nonzero ρ would not show that faster drinks become popular; source confounding is likely.*

Statistical evidence. *ρ = -0.290, p = < 0.0001; CI [-0.3173572925706958, -0.26082386797614776].*

Caveat. *Complete cases only; prep and votes are both missing for many rows.*

Figure 31. Preparation time by top 10 categories (boxplot).



What this figure shows. *Boxplots of parsed minutes in the ten most frequent non-missing categories (n = 629 recipes with both category and prep time). The vertical axis is clipped near the 99th percentile (27.2 min) so a few extreme infusions do not compress the boxes.*

Key insight. *Medians remain short in every leading category (typical medians Fruity Cocktails: 5.0 min, Tequila Cocktails: 3.0 min, Martinis: 5.0 min, Margaritas: 5.0 min). Outliers exist in several classes.*

Why it matters. *Category-level prep time is the comparison the extended EDA notebook emphasized and that this report previously omitted. Categories are the ten most frequent among recipes that have both Category and parsed prep time (not the global top-10 labels, because category and prep time rarely co-occur).*

Statistical evidence. *Top categories = Fruity Cocktails, Tequila Cocktails, Martinis, Margaritas, Gin Drinks, Rum Cocktails, Champagne Drinks, Fall Cocktails, Whiskey Cocktails, Gin Cocktails. Per-category n = {'Fruity Cocktails': 108, 'Tequila Cocktails': 67, 'Martinis': 65, 'Margaritas': 61, 'Gin Drinks': 57, 'Rum Cocktails': 56, 'Champagne Drinks': 55, 'Fall Cocktails': 54, 'Whiskey Cocktails': 53, 'Gin Cocktails': 53}.*

Caveat. *Do not treat these ten bars as a global ranking of cocktail types; Category coverage is incomplete and the y-cap hides the most extreme points.*

5.6 Nutritional characteristics

Nutrient strings are present for 5,694 / 16,895 recipes (66.3% missing). Two formats occur: compact “184 Calories|3g Fat|8g Carbs” and colon-style “Calories: 239 kcal|Carbohydrates: 14 g|…”. Carbs and Carbohydrates were mapped to one carbohydrates variable after checking that both are gram masses in the observed text. The file does not state per-serving standardization. Completeness and moments for each extracted nutrient are in §6.4 and the appendix tables. PCA, if performed, uses complete cases only.

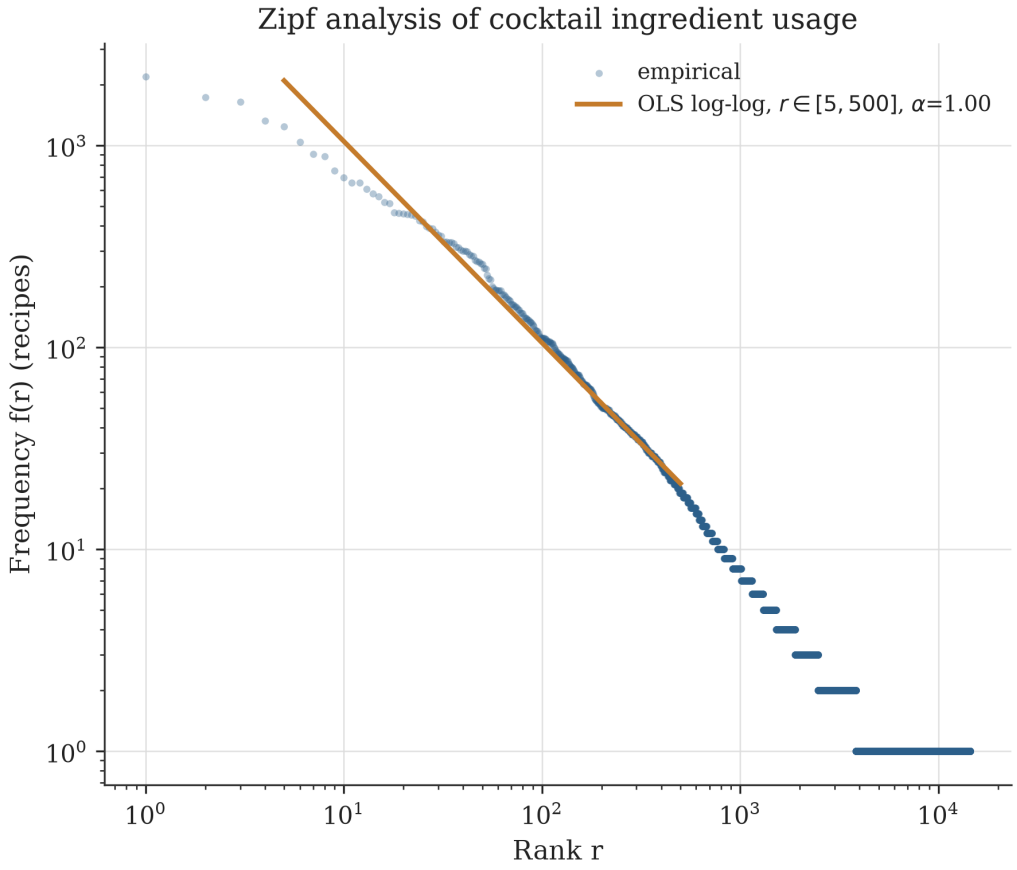
6. Statistical Laws in Cocktail Recipes

6.1 Zipf analysis

Model: f(r) ~ r^{-α} on ingredient ranks. Primary descriptive fit is log–log OLS on r ∈ [5, 500], giving α = 0.998, R^2 = 0.991, n_points = 496. The unrestricted full-range OLS α = 0.878 with R^2 = 0.884, which is a reminder that curvature at the head and far tail moves the exponent. Following Clauset et al. as in the paper’s Methods, a discrete power law was also fitted to the vector of frequencies (P(f) ~ f^{-α_freq}), which is not identical to the rank–frequency exponent. MLE α_freq = 1.848, x_min = 2.00, KS = 0.356, n_tail = 3840. Likelihood-ratio comparisons versus alternatives: lognormal: R=-3.99, p=6.61e-05; exponential: R=14.66, p=1.22e-48. Positive R favors the power law in the Alstott implementation; p-values near 1 do not automatically validate a power law over the whole range. A heavy tail in cocktails means a small core bar (spirits, citrus,

sugar, bitters, ice) is reused constantly, while a long tail of liqueurs, garnishes, and branded products supplies diversity. The fit is reported as Zipf-like, not as a proven unique generative law.

Figure 16. Empirical rank–frequency distribution and fitted Zipf relation.



What this figure shows. *Points: all ingredients. Line: OLS log–log fit on the primary rank window.*

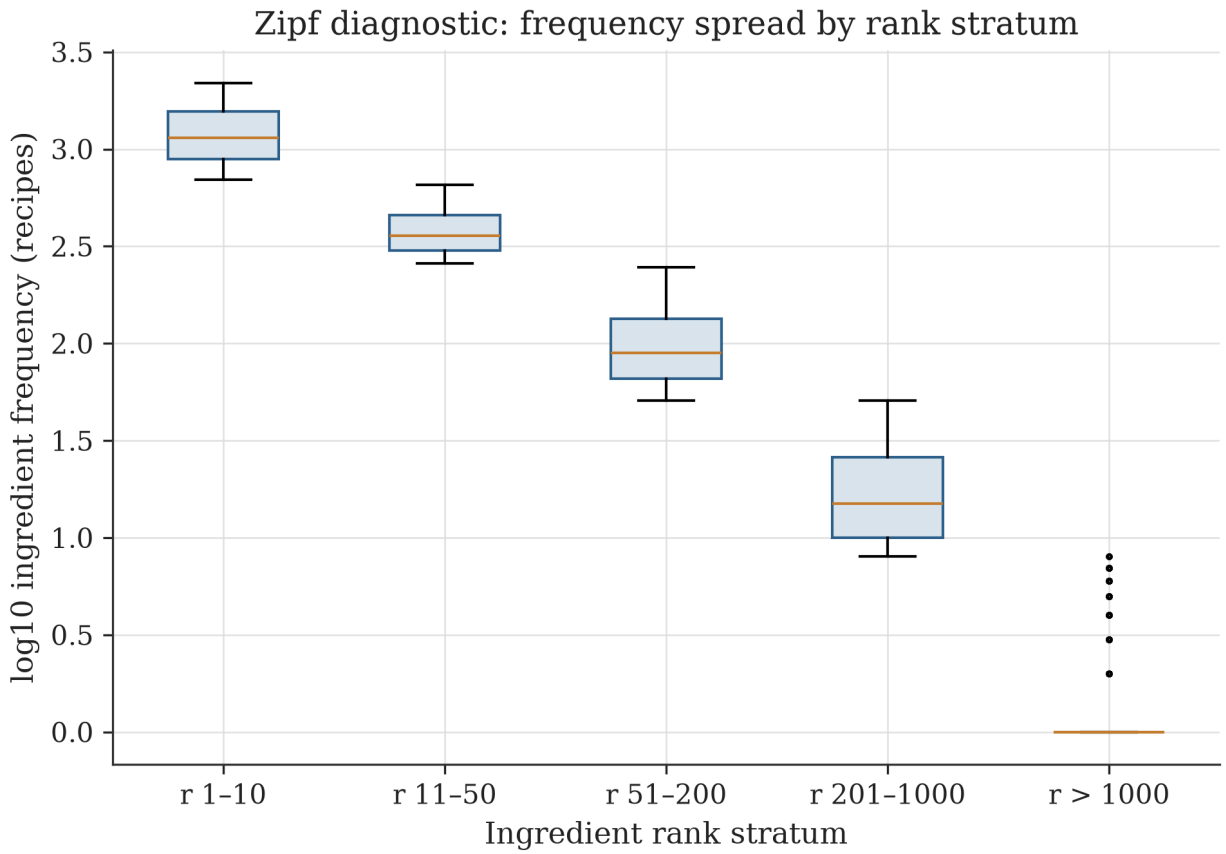
Key insight. *$\alpha \approx 1.00$ on $r \in [5, 500]$. The empirical curve is heavy-tailed but not a perfect straight line.*

Why it matters. *If a few ingredients dominate, cocktail "vocabulary" is reusable and predictable; the tail still allows innovation.*

Statistical evidence. *OLS $\alpha = 0.998$, $R^2 = 0.991$, $n = 496$. Clauset $\alpha_{freq} = 1.848$, xmin = 2.00.*

Caveat. *R^2 is descriptive. Range sensitivity is tabulated in zipf_range_sensitivity.csv. Do not equate α with the paper's 1.53 without noting corpus differences.*

Figure 32. Zipf diagnostic: log-frequency boxplots by rank stratum.



What this figure shows. *Each box is the distribution of log10 recipe-frequency for ingredients in a rank window (1–10, 11–50, 51–200, 201–1000, >1000).*

Key insight. *Median log-frequency falls across strata (r 1–10: 3.06, r 11–50: 2.55, r 51–200: 1.95, r 201–1000: 1.18, r > 1000: 0.00), which is the boxplot view of the same heavy tail shown on the log–log rank plot.*

Why it matters. *This is a compact way to see that a handful of head ingredients sit far above a long body of rare items, without claiming a unique power-law exponent.*

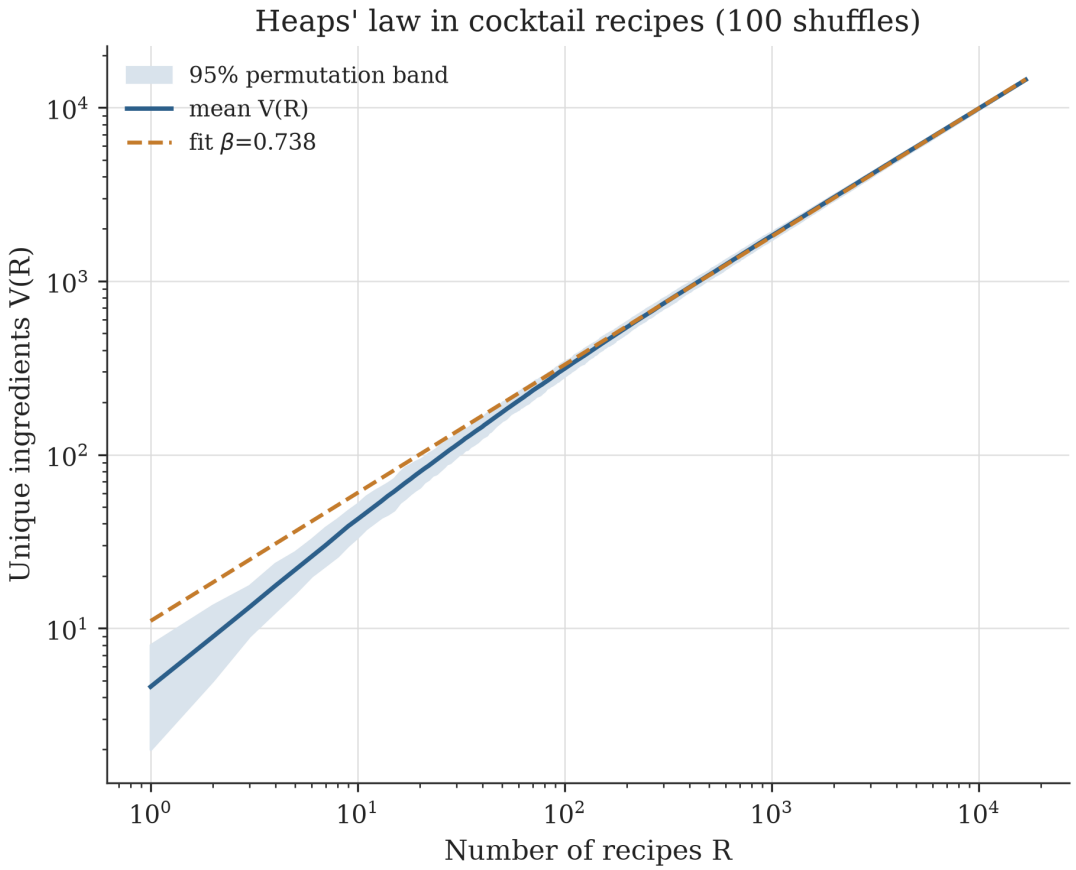
Statistical evidence. *Stratum sizes n = {'r 1–10': 10, 'r 11–50': 40, 'r 51–200': 150, 'r 201–1000': 800, 'r > 1000': 13466}.*

Caveat. *Rank bins are analyst choices. They illustrate Zipf-like decay; they are not a substitute for the MLE/OLS fits in Figure 16.*

6.2 Heaps analysis

V(R) is the number of distinct standard-normalized ingredients after R recipes. 100 random permutations were used. Mean β = 0.7385 (SD 0.0066; 95% permutation interval [0.7254, 0.7515]); mean log–log R^2 = 0.9996. $\beta < 1$ is sublinear growth: new recipes keep introducing ingredients, but the marginal discovery rate declines. The paper's global $\beta \approx 0.56$ is a reference point, not a target. Cocktail $\beta \approx 0.74$ can differ because of domain (drinks vs meals), corpus size (16,872 vs 118,083), normalization, and source mix. Sensitivity: moderate β = 0.720; core β = 0.719.

Figure 17. Heaps curve with permutation band.



What this figure shows. *Mean V(R) and 95% band across 100 shuffles, log–log axes, with a power-law overlay.*

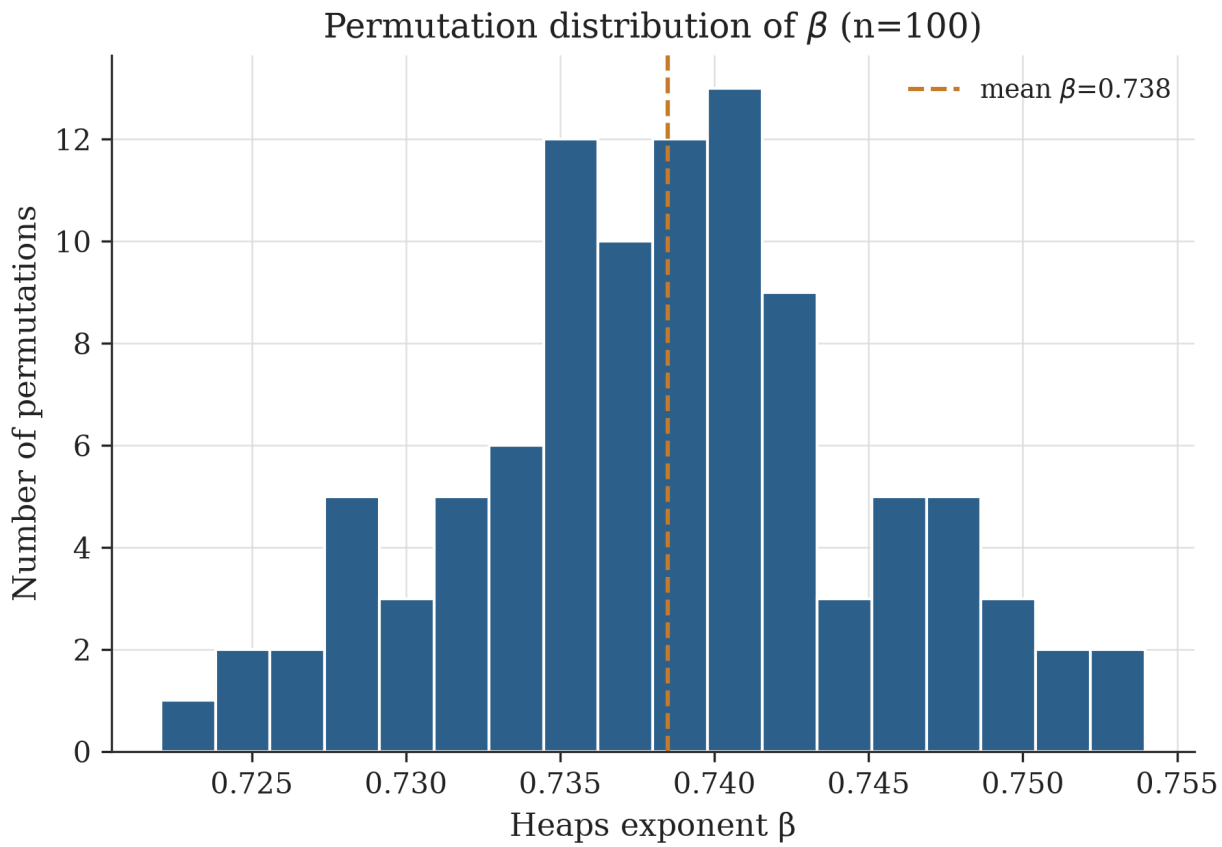
Key insight. *Growth is sublinear (β ≈ 0.738); the band is tight, so ordering noise is small relative to the trend.*

Why it matters. *Cocktail innovation continues (the curve still rises at the corpus end) but with diminishing new-ingredient returns.*

Statistical evidence. *β_mean = 0.7385; 95% interval [0.7254, 0.7515]; final mean V = 14,466; n_recipes = 16,872.*

Caveat. *Fits use $R \geq 20$ to reduce the discrete start-up; K is a nuisance scale parameter.*

Figure 18. Distribution of β across permutations.



What this figure shows. *Histogram of log–log OLS exponents, one per shuffle.*

Key insight. *β estimates concentrate tightly around 0.738 (SD 0.0066).*

Why it matters. *The Heaps exponent is a stable corpus property, not an artifact of one recipe order.*

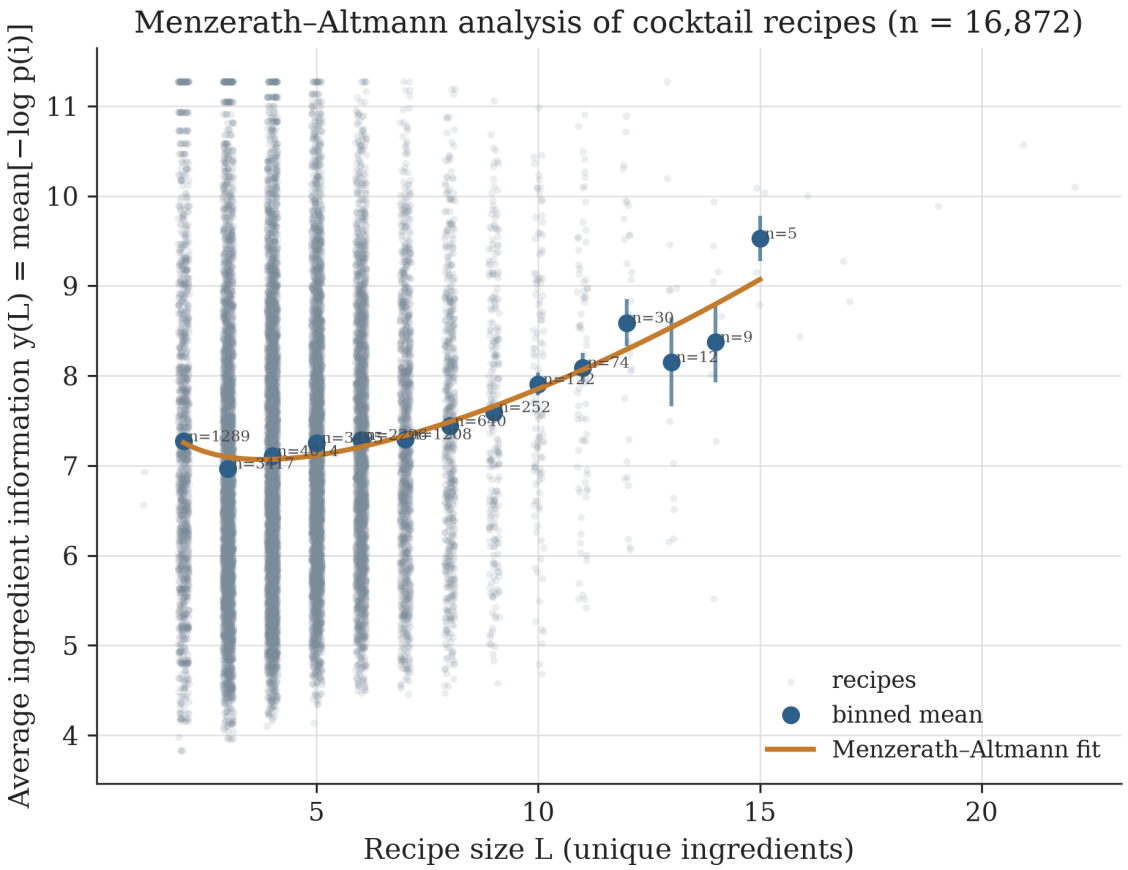
Statistical evidence. *n_shuffle = 100; mean R^2 = 0.9996.*

Caveat. *Least-squares on log V vs log R is the paper's estimator; it is not maximum likelihood.*

6.3 Menzerath–Altmann analysis

Following the paper, p(i) = f_i / Σ f_j, s_i = –log p(i), L = unique ingredients, y = mean s_i in the recipe. On standard normalization, n = 16,872, Spearman(L, y) = 0.101. Binned MA fit y(L)=a L^b exp(cL): a = 7.446, b = -0.157, c = 0.042, R^2 = 0.898, n_bins = 14. The paper's global fit (a=13.16, b=–0.26, c=–0.019, R^2=0.99, minimum near L≈13) lives on a length range cocktails barely populate. If y(L) is flat, increasing, or poorly fitted, that is a result: short drink recipes may not exhibit the same complexity–length trade-off as 10-ingredient meals. Sensitivity (Spearman): moderate 0.066; core 0.051; p(i)=f_i/N_recipes 0.101.

Figure 19. Average ingredient information versus recipe size.



What this figure shows. *Grey points: recipes. Blue: binned means with SE bars and bin counts. Orange: MA functional fit on bins with n≥5.*

Key insight. *Spearman ρ(L, y) = 0.101. The MA R^2 on bins is 0.898.*

Why it matters. *A decreasing y(L) would mean longer drinks pad with common ingredients; an increasing or flat curve would differ from the paper's efficiency–expressivity story.*

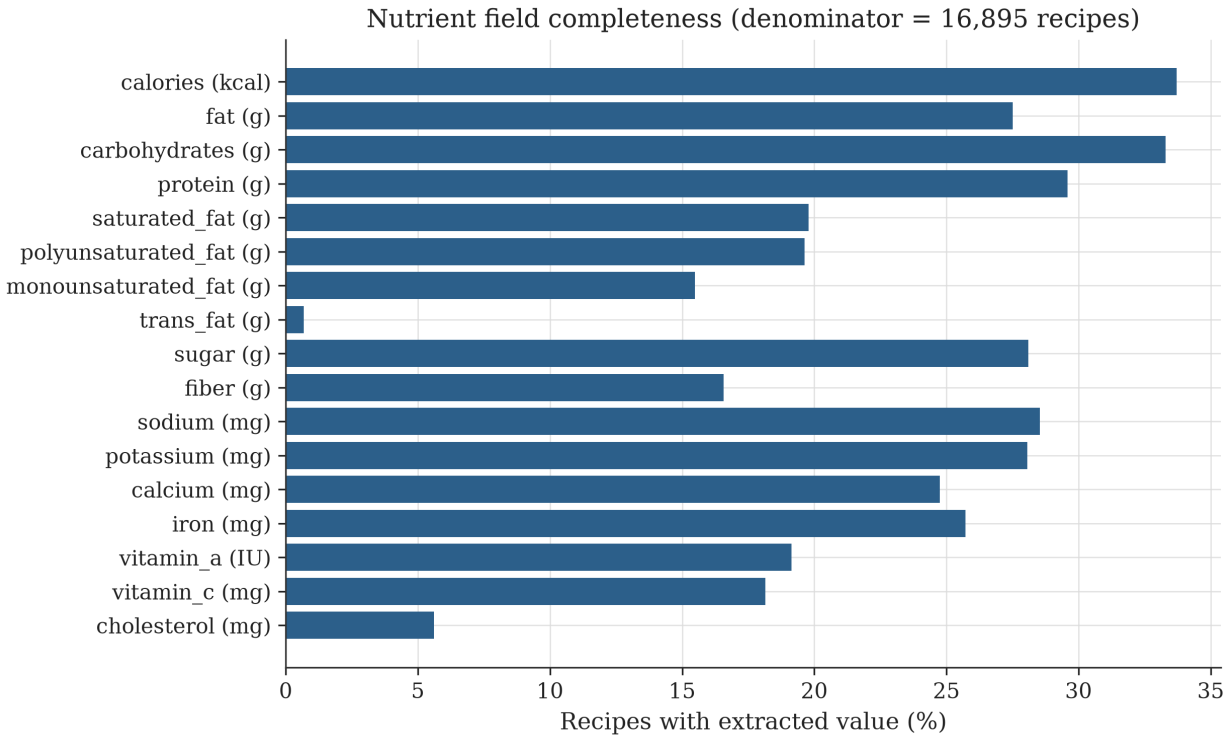
Statistical evidence. *n_recipes = 16,872; L range 1–22 (median 4.0). a,b,c = 7.45, -0.16, 0.042; R^2 = 0.898.*

Caveat. *Few recipes have L near 13, so a turning point at the paper's scale cannot be identified even if the functional form is forced.*

6.4 Nutrient distribution analysis

Positive-valued extracted nutrients were compared to log-normal, gamma, Weibull, Gaussian, and exponential laws via KS distance (smaller is better). Calories: n+ = 5692, log-skewness = -1.084, best = lognorm. Carbohydrates: n+ = 5614, log-skewness = -1.551, best = gamma. The paper reported mean KS 0.0727 for log-normal on meal macronutrients. Cocktail nutrients are sparser, possibly not per serving, and include zeros (especially protein/fat) that are excluded from log tests. Log-normality is supported only where KS and log-skewness jointly agree—not because a histogram looks bell-shaped.

Figure 22. Nutrient completeness.



What this figure shows. *Percent of all 16,895 recipes with a parsed value for each nutrient.*

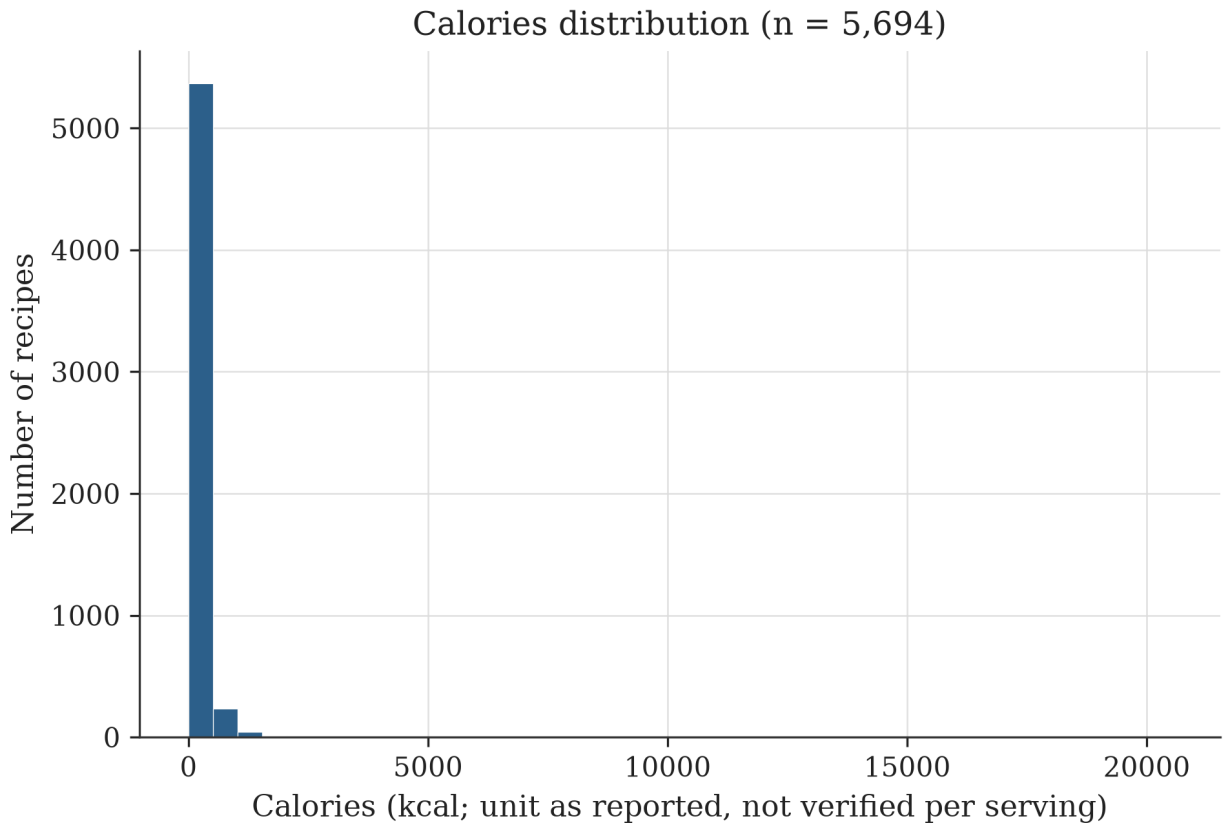
Key insight. *Even calories, the most common field, cover only a minority of the full table; micronutrients are rarer.*

Why it matters. *Nutrient EDA is a study of the nutrient-reporting subset, which may be biased toward recipe sites that compute nutrition panels.*

Statistical evidence. *Nutrient strings present: 5,694 (33.7% of rows have a non-null Nutrients field).*

Caveat. *A non-null string can still omit individual nutrients.*

Figure 23. Calories distribution.



What this figure shows. *Histogram of extracted calorie values as reported.*

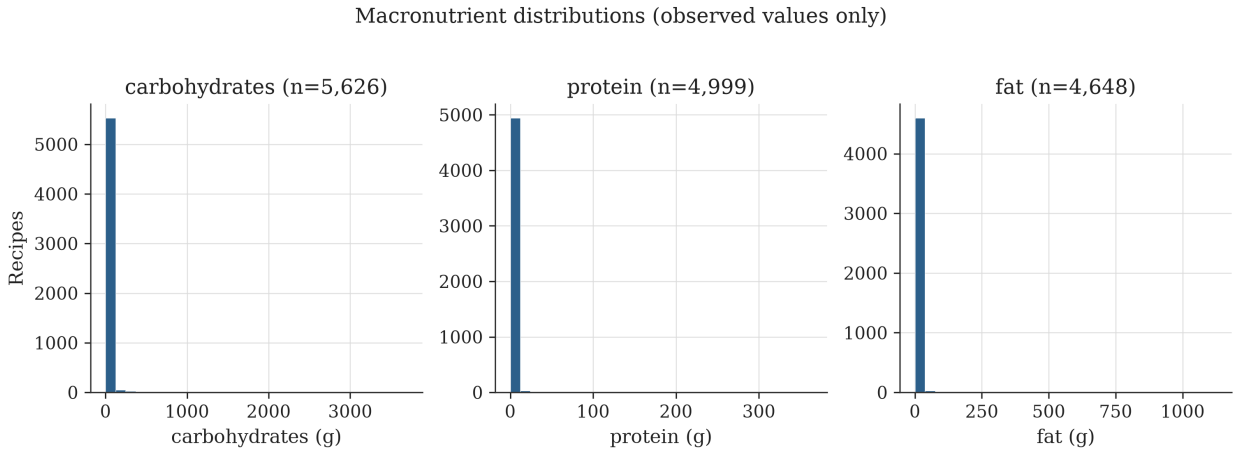
Key insight. *The distribution is right-skewed with extreme high values that may reflect pitchers, multiple servings, or parsing/source errors—not typical single-glass cocktails.*

Why it matters. *Calories cannot be treated as a standardized per-drink nutrient without serving metadata.*

Statistical evidence. *See nutrient_descriptives.csv for n, mean, median, SD, IQR, min, max.*

Caveat. *Zeros and missing are excluded from this histogram if non-finite.*

Figure 24. Macronutrient distributions.



What this figure shows. *Side-by-side histograms of carbohydrates, protein, and fat in grams where extracted.*

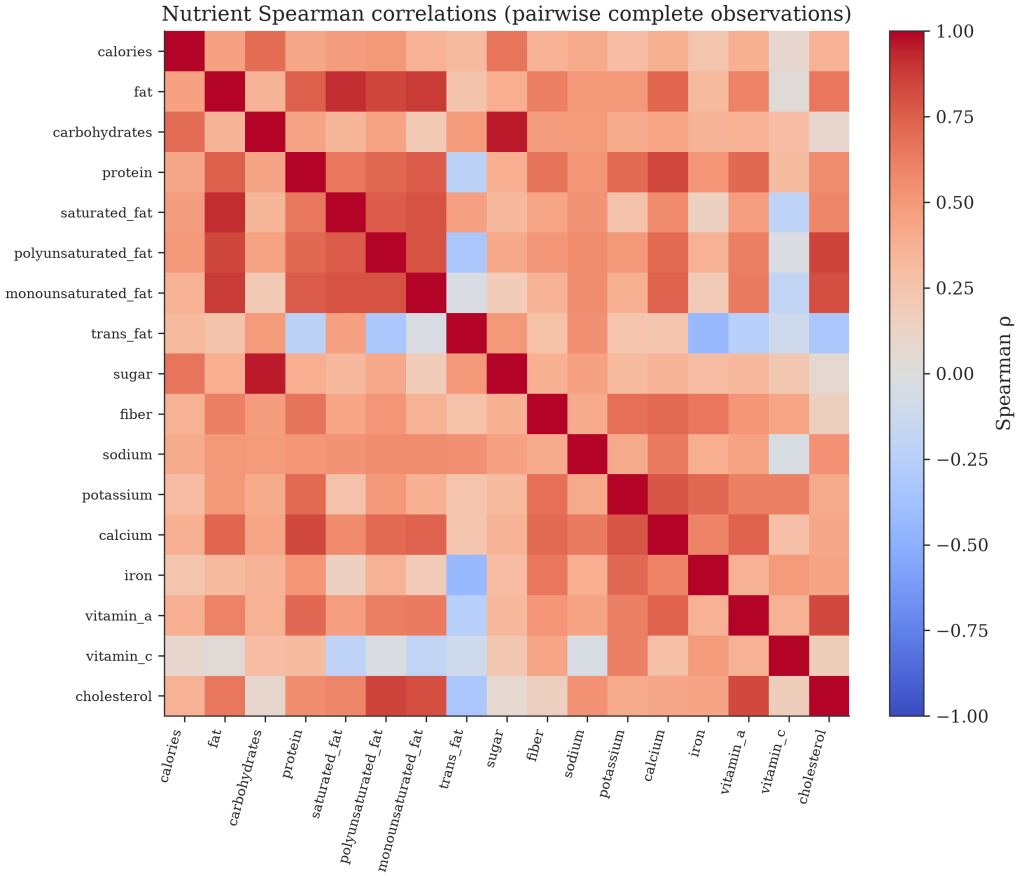
Key insight. *Protein is often near zero in drinks; carbohydrates (sugars/juices) and occasional fats (cream, egg, dairy liqueurs) differ sharply.*

Why it matters. *Cocktail macronutrient structure is not interchangeable with cooked-meal macronutrients in the paper.*

Statistical evidence. *Sample sizes differ by nutrient; see Figure 22 and descriptives table.*

Caveat. *Units are grams as parsed; cream-based drinks drive fat tails.*

Figure 25. Nutrient Spearman correlation matrix.



What this figure shows. *Pairwise complete-case Spearman correlations among nutrients with ≥80 observations.*

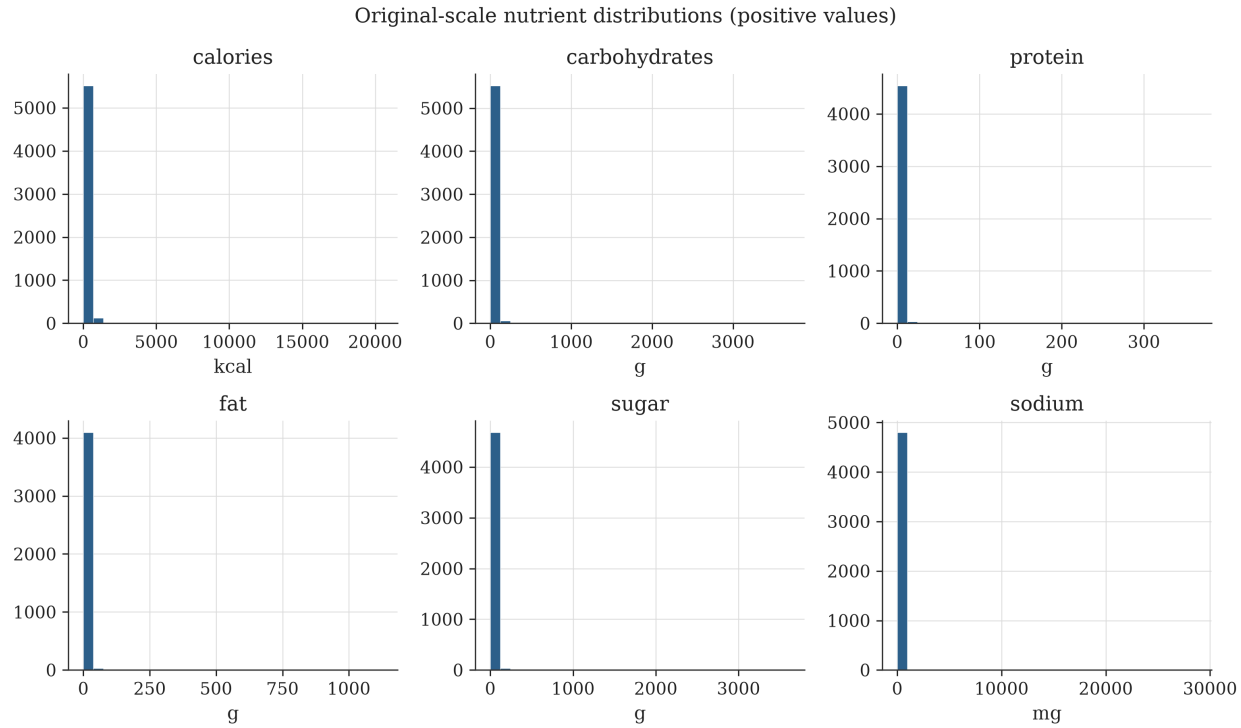
Key insight. *Positive associations among energy-related fields (calories, carbohydrates, sugar, fat) are expected arithmetically if they co-occur on the same panel, not as a biological law.*

Why it matters. *Correlations are descriptive of reported panels.*

Statistical evidence. *Spearman chosen because of skew and outliers.*

Caveat. *Pairwise deletion means each cell has its own n.*

Figure 26. Original-scale nutrient distributions.



What this figure shows. *Positive observations for calories, carbohydrates, protein, fat, sugar, and sodium when n+ is sufficient.*

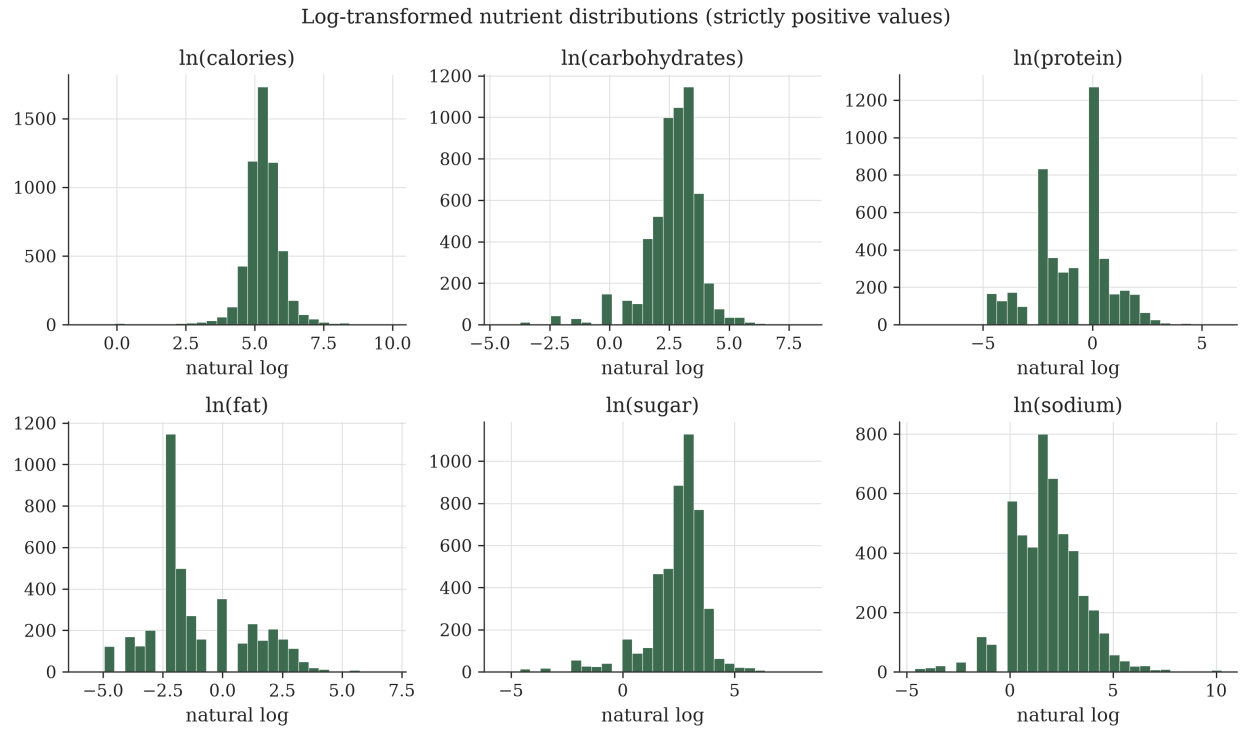
Key insight. *All are right-skewed on the original scale.*

Why it matters. *Skew is why the paper examines log space.*

Statistical evidence. *n+ listed in 08_nutrition.json per nutrient.*

Caveat. *Zeros excluded.*

Figure 27. Log-transformed nutrient distributions.



What this figure shows. *Natural logs of strictly positive values.*

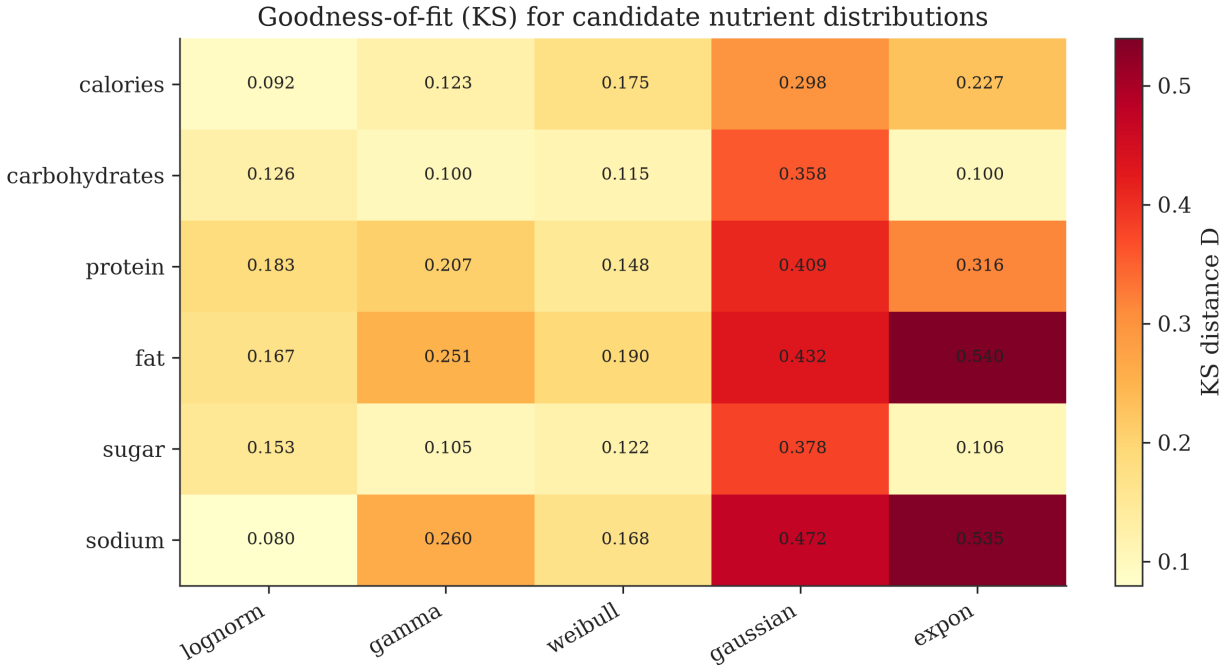
Key insight. *Log-skewness near 0 would support log-normal symmetry; observed values include calories -1.084 and carbohydrates -1.551.*

Why it matters. *Translational invariance across macronutrients (paper Fig. 3b) is only weakly testable with three incomplete cocktail macros.*

Statistical evidence. *Log transform applied only for x>0.*

Caveat. *Visual unimodality is not a KS test.*

Figure 28. Distribution-fit comparison (KS distances).



What this figure shows. *Rows: nutrients. Columns: candidate laws. Cell text is KS D.*

Key insight. *Best fits by nutrient: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm. Log-normal is not automatically best for every cocktail nutrient.*

Why it matters. *This is the paper's GOF logic applied honestly to drinks data.*

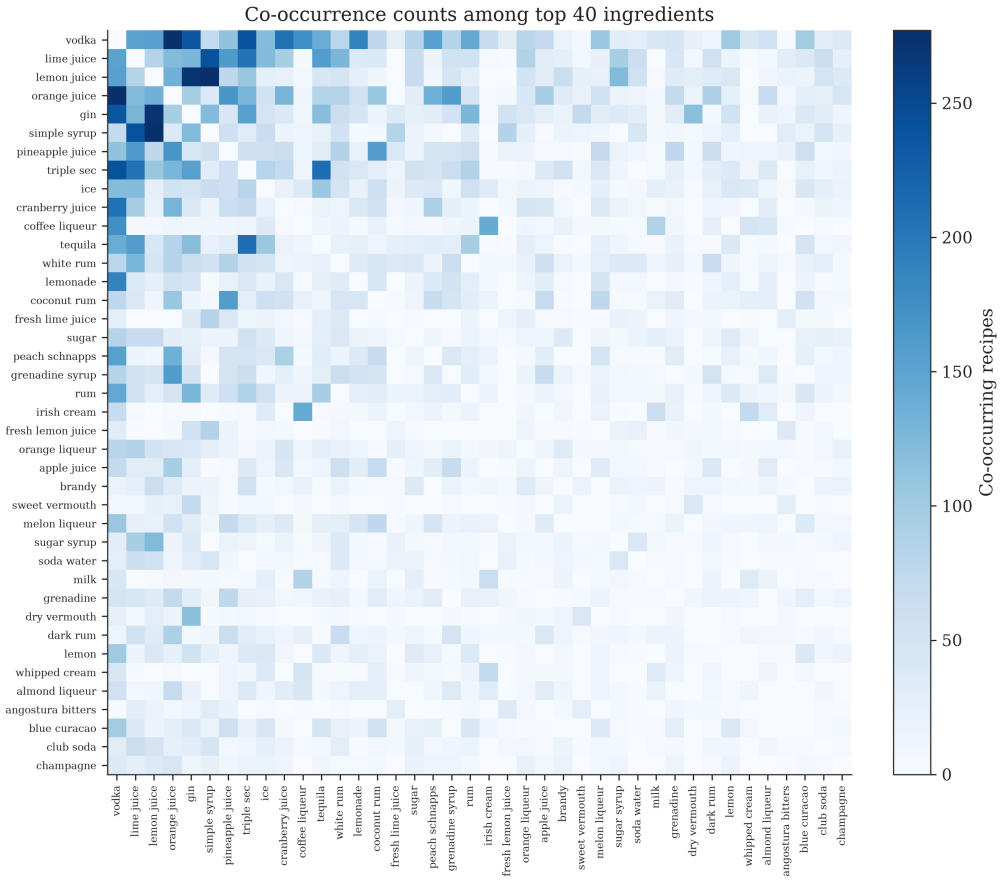
Statistical evidence. *D is the KS statistic on positive observations with floc=0 for skewed candidates where used.*

Caveat. *KS on large n flags tiny deviations; ranking of D is more relevant than a binary p-value.*

7. Ingredient Co-occurrence

Beyond the paper, co-occurrence was counted among the top 40 ingredients. PMI uses p(i,j)/(p(i)p(j)) with a minimum co-occurrence of 25 recipes to limit rare-event inflation. The network retains positive PMI edges (n_edges = 166). Clusters reflect recipe conventions (e.g., spirit+citrus+sweetener, bitters+sugar, cream+chocolate liqueurs) rather than measured flavor-compound pairing. Co-occurrence is not causality and is not food-pairing chemistry.

Figure 20. Co-occurrence heatmap (top ingredients).



What this figure shows. *Cell (i,j) is the number of recipes containing both ingredients among the top 40.*

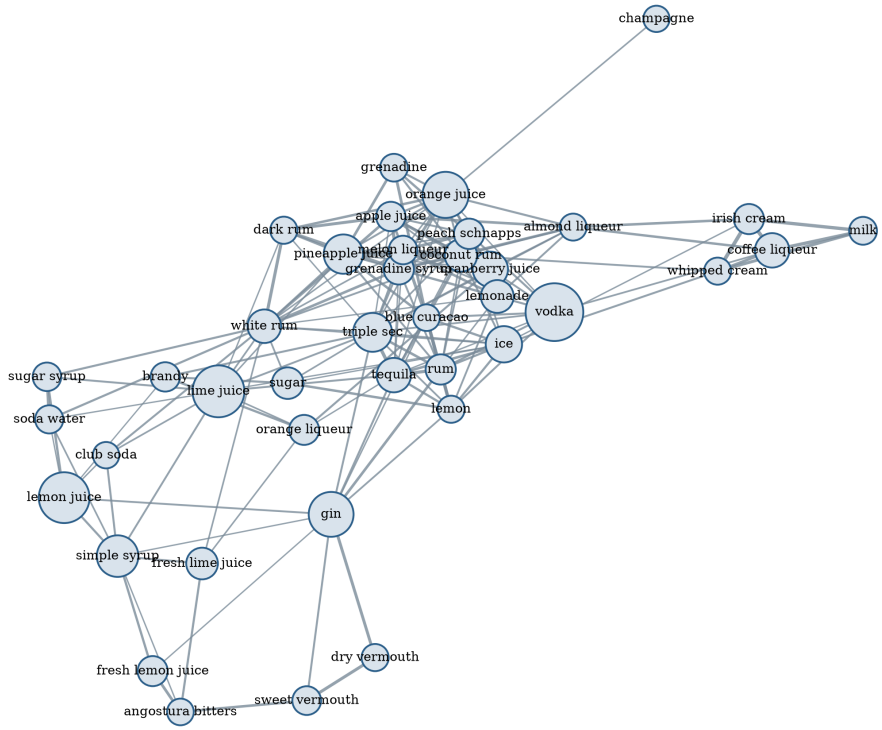
Key insight. *Darker cells mark bar-standard pairs (citrus with spirits; sugar with citrus; ice with many bases).*

Why it matters. *These pairs describe recipe incidence, the combinatorial skeleton of the cocktail corpus.*

Statistical evidence. *n_recipes in long table = 16,872.*

Caveat. *Top-N selection conditions on frequency, so rare but culturally important pairs are absent.*

Figure 21. Positive-PMI association network.



What this figure shows. *Nodes: top 40 ingredients (size ~ frequency). Edges: co-occurrence ≥ 25 and PMI > 0.4.*

Key insight. *Highest-PMI edges include: irish cream–whipped cream (PMI=2.09), coffee liqueur–irish cream (PMI=2.08), lemon–blue curacao (PMI=1.87), irish cream–milk (PMI=1.83), triple sec–tequila (PMI=1.82), coffee liqueur–milk (PMI=1.82).*

Why it matters. *Network clusters are usage cliques, useful for generating hypotheses, not for asserting compatibility.*

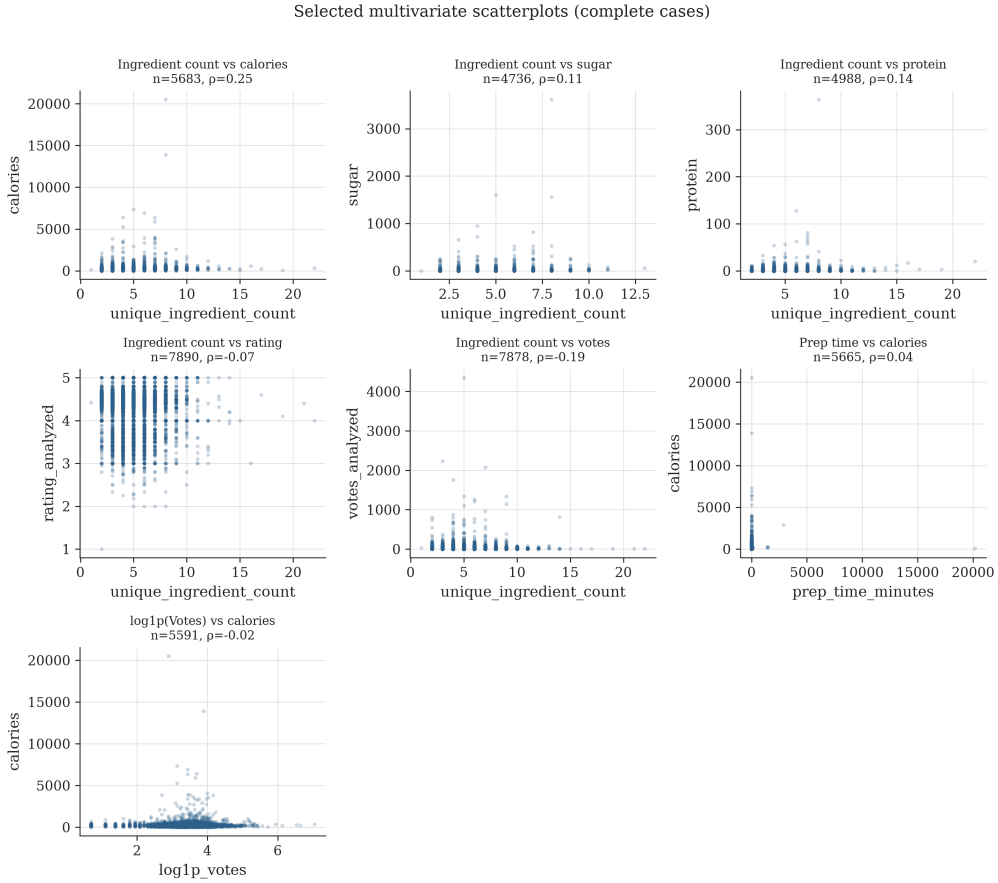
Statistical evidence. *n_edges = 166.*

Caveat. *Thresholds are analyst choices; they hide weak ties.*

8. Multivariate Relationships

Pre-specified pairs linked complexity (ingredient count, prep time), popularity (votes, ratings), and nutrients (calories, sugar, protein). Spearman correlations with bootstrap CIs are stored in multivariate_spearman.csv. Largest |ρ| among those pairs: ingredient count vs calories (ρ=0.25, n=5683), ingredient count vs votes (ρ=-0.19, n=7878), ingredient count vs protein (ρ=0.14, n=4988). Effect sizes are interpreted over p-values because n can be large in some complete-case slices and tiny in nutrient slices. No pair is treated as a causal graph.

Figure. Selected multivariate scatterplots.



What this figure shows. *Complete-case scatters for ingredient count, calories, sugar, protein, rating, votes, and prep time.*

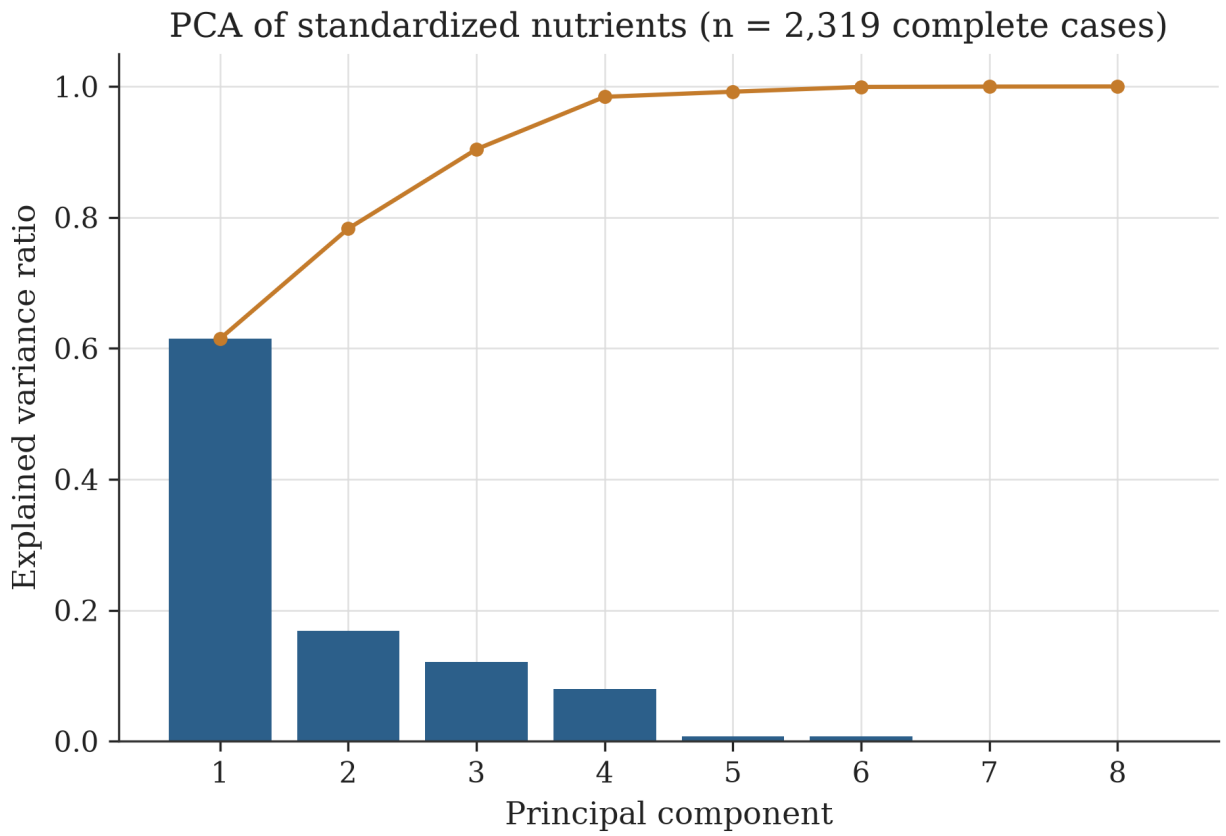
Key insight. *Visual structure is weak to moderate; nutrient axes are dominated by skew and a smaller n.*

Why it matters. *Cocktail 'complexity' is not a strong linear predictor of reported nutrition or popularity in this file.*

Statistical evidence. *See multivariate_spearman.csv; strongest listed above.*

Caveat. *Each panel has its own complete-case n.*

Figure 29. PCA explained variance.



What this figure shows. *Scree of standardized nutrients on n = 2,319 complete cases, variables = ['calories', 'fat', 'carbohydrates', 'protein', 'sugar', 'fiber', 'sodium', 'saturated_fat'].*

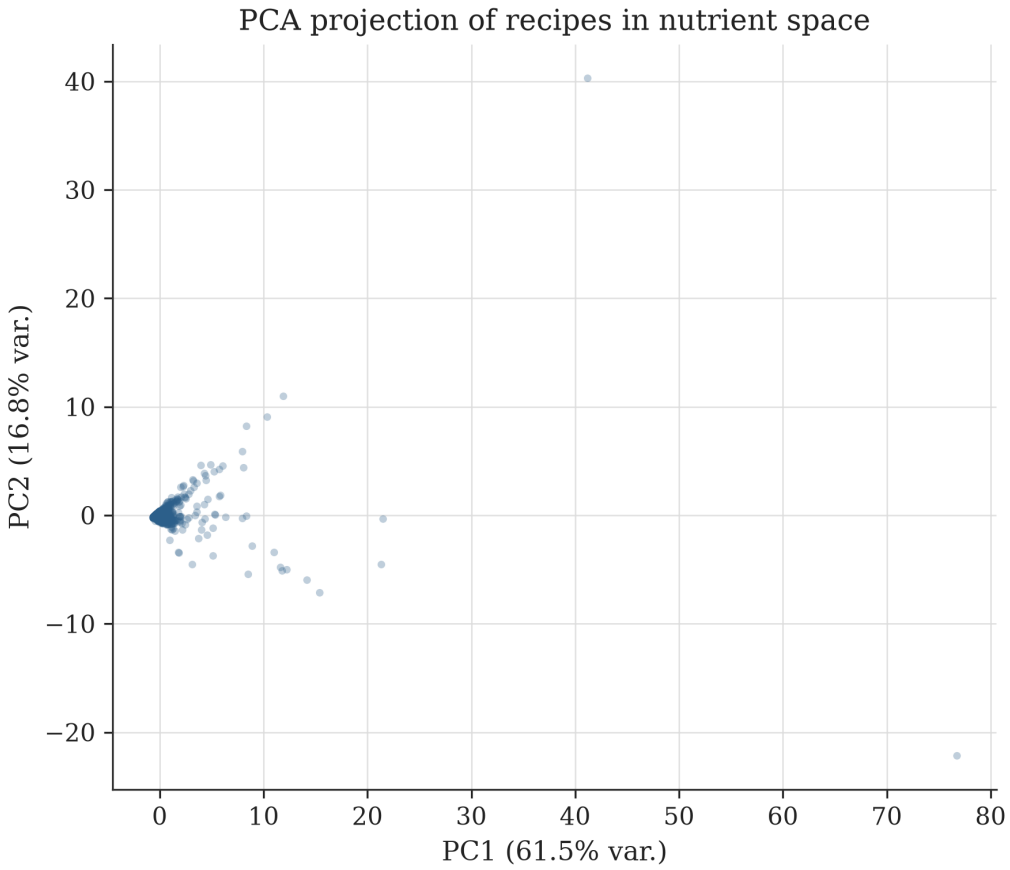
Key insight. *PC1 explains 61.5% and PC2 16.8%.*

Why it matters. *PCA is a descriptive compression of co-reported nutrient panels, not a latent 'health' index.*

Statistical evidence. *Cumulative variance in first two PCs = 78.3%.*

Caveat. *Complete-case selection biases toward richer nutrition labels.*

Figure 30. PCA projection.



What this figure shows. *Each point is a recipe in the complete nutrient subset.*

Key insight. *Spread along PC1 typically aligns with overall magnitude (calories/macros) if loadings share sign.*

Why it matters. *PC1 loadings: {'calories': 0.4308058376461811, 'fat': 0.3910828444355299, 'carbohydrates': 0.3655308884951331, 'protein': 0.40766508205135343, 'sugar': 0.3545004777225405, 'fiber': 0.28136545080174763, 'sodium': 0.08351134622978185, 'saturated_fat': 0.3870990040166974}.*

Statistical evidence. *n = 2,319.*

Caveat. *Do not cluster-interpret without a pre-registered clustering analysis (not performed).*

9. Comparison with the Reference Paper

Comparison with Universal Statistical Laws Governing Culinary Design. The paper is a methodological reference. CocktailDB is not required to match its exponents.

Phenomenon	Paper	This dataset	Method	Evidence	Interpretation
Zipf	f(r)~r^{-α}, α_global≈1.53; cuisine mean≈1.39; Clauset MLE on frequencies; LR vs log-normal/exponential	OLS α≈1.00 on r=5–500; Clauset α_freq≈1.85, xmin=2.0	Rank–frequency OLS + Clauset MLE	Heavy tail yes; perfect power law not assumed	Partially consistent: Zipf-like reuse in a smaller drink vocabulary
Heaps	V(R)~R^β, β≈0.56 (100 shuffles; methods also mention 10 for error bars)	β≈0.74 (100 shuffles; 95% [0.73, 0.75])	Permutation + log–log OLS	mean R²=1.000	Consistent in sign (sublinear); similar magnitude is coincidental unless replicated on other drink corpora
Menzerath–Altmann	y(L)=a L^b e^{cL}; a=13.16,b=−0.26,c=−0.019; R²=0.99; min L≈13	ρ(L,y)=0.10; binned R²=0.90; L median=4.0	Information s_i=−log p(i); NLS	Short L range	Different / weakly testable at the paper's length scale

Phenomenon	Paper	This dataset	Method	Evidence	Interpretation
Nutrient distributions	Log-normal macros; mean KS 0.073 vs worse alternatives	Best KS by nutrient: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm; incomplete coverage	KS on x>0	Nutrient n much smaller; serving unknown	Partially testable; not a reproduction of meal log-normality
Ingredient diversity	19,019 unique ingredients; 1.15M mentions	14,466 standard unique; hapax=73.5%	Normalized tokens	Smaller closed bar vocabulary	Consistent with constrained vocabulary, different scale
Recipe complexity	Unimodal 8–12 ingredients; mean ~10; ~9 steps	median L=4; mean=4.7; no steps	Unique ingredient count	No technique/step data	Different: cocktails are shorter; steps not testable

10. Major Findings

1. Ingredient usage is concentrated: the most common normalized ingredient is “vodka” (2,191 recipes; 13.0% of the corpus). 73.5% of the vocabulary occurs in only one recipe.

2. Rank–frequency curves are heavy-tailed. A log–log OLS Zipf exponent of $\alpha \approx 1.00$ was obtained on ranks 5–500 (descriptive $R^2 = 0.991$). A Clauset-style power-law fit to the frequency distribution is reported separately and is not treated as automatic confirmation of Zipf’s law.

3. Heaps’ law is clearly sublinear: mean $\beta = 0.738$ (100 permutations; 95% interval [0.725, 0.751]). This is close to the paper’s global $\beta \approx 0.56$ but is estimated on a smaller, cocktail-only corpus and is not a target to match.

4. Menzerath–Altmann: Spearman correlation between recipe size L and mean ingredient information y(L) is $\rho = 0.101$. The paper’s decreasing-then-rising global pattern is not assumed; the fitted MA parameters and R^2 are reported in §6.3.

5. Recipe size is small relative to the paper’s food corpus: mean unique ingredients = 4.67 (median 4.0; n = 16,872).

6. Ratings, where observed, are high and left-censored toward 5 stars (mean 4.32, median 4.45, n = 7,890). All 116 Rating=0 rows have Votes=0 and were treated as unrated.

7. Votes are heavy-tailed. Spearman association of votes with rating is $\rho = -0.058$ (n = 7878). This is an association, not evidence that votes cause ratings.

8. Cuisine cannot support cuisine-level laws: only 393 non-missing values (2.33% of the frequency corpus), with labels that mix drink types and nationalities.

9. Category is available for 6,925 recipes (41.0% of the frequency corpus); category comparisons are conditional on that subset.

10. Nutrient strings exist for 5,694 recipes (66.3% missing). Log-space KS comparisons are reported per nutrient; log-normality is not claimed globally. Best KS candidates include: calories: lognorm; carbohydrates: gamma; protein: weibull; fat: lognorm; sugar: gamma; sodium: lognorm.

11. Strongest complete-case Spearman associations among the pre-specified multivariate pairs: ingredient count vs calories (ρ=0.25, n=5683), ingredient count vs votes (ρ=-0.19, n=7878), ingredient count vs protein (ρ=0.14, n=4988).

12. Cooking techniques, utensils, instruction steps, and 26-cuisine stratification from the reference paper cannot be reproduced because those variables are absent.

11. Limitations

1. Missing values are extensive and structured; every analysis reports its own n.

2. Cuisine is too sparse and semantically inconsistent (Cocktail vs American vs Vegan) for cuisine-level laws.

3. Category incompleteness (about 59% missing) blocks globally representative category inference.

4. Nutrient fields mix compact and colon formats; units were harmonized only where the text supported it.

5. Ingredient normalization is partly subjective; sensitivity across four levels is provided but residual quantity/brand strings remain.

6. The file likely concatenates heterogeneous sources (rating sites, recipe blogs, nutrition-labelled pages).

7. Duplicate names may be variants or competing scrapes; only exact row and Name+Ingredients clones were dropped.

8. Sampling bias: web cocktails over-represent English-language, Instagram-era, and database-popular drinks.

9. Votes measure attention, not quality.

10. Ratings with few votes are unstable and piled at 5.

11. Observational correlations are not causal.

12. Cocktail-specific composition may not generalize to global cuisine—the paper's claim domain.

13. Corpus size (16,872 frequency recipes) is an order of magnitude below 118,083.

14. No reliable geographic or cultural metadata.

15. Prep times and nutrients may mix servings, infusions, and panel-generator artifacts; extremes were investigated, not auto-deleted.

12. Future Work

A cocktail-specific NER pass would align the corpus more closely with the paper's structured entities. Linking drinks to flavor-compound data would test whether co-occurrence clusters match chemistry. Serving-size standardization is required before nutrient log-normality can be compared to packaged-food or RecipeDB results. Larger multilingual bar corpora would show whether β and α are drink-domain constants. Historical menus could separate traditional formulas from modern riffs.

13. Conclusion

CocktailDB supports the paper's high-level claim that culinary compositions reuse a core vocabulary and accumulate novelty sublinearly: Zipf-like rank–frequency structure and Heaps $\beta \approx 0.74$ are visible after careful normalization. It does not support a blanket transfer of the global Menzerath–Altmann minimum at $L \approx 13$, cuisine-universal exponents, technique scaling, or fully log-normal meal macronutrients. The honest result is mixed—and that is the scientifically relevant one.

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Appendix

A. Reproducibility. Run python scripts/run_all.py from eda_project/. Numerical results live in tables/json/. Figures are written to figures/ as PNG and PDF. This report reads those files and does not hard-code exponents.

B. Duplicate policy. Exact rows: remove extras (no information). Name+Ingredients clones: remove extras for frequency (avoid double-counting the same formula). Same name, different ingredients: keep (variants).

C. Paper Methods notes used here. Ingredient frequency = recipe incidence; Zipf via rank–frequency plus Clauset MLE on frequencies with xmin by KS; Heaps $V(R)=K R^{\beta}$ by log–log OLS on permutations; MA $y(L)=a L^{b} \exp(cL)$ by nonlinear least squares; nutrients: log transform of positives, log-skewness, KS versus log-normal/gamma/Weibull/Gaussian/exponential.

D. Frequency corpus size = 16,872. Raw rows = 16,895. Unique names = 14,916. Unrated zeros = 116.